

PE-EC802B: Industrial Automation and Control

Study Guide and Comprehensive Exam Preparation Notes

Semester VIII B.Tech ECE (MAKAUT)
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Chapter 1

Sensors, Actuators & Signal Conditioning

1.1 Section 1: Industrial Sensors [15M] [Priority: High]

1.1.1 1. Transducers: Active vs. Passive

A **transducer** is a physical device that converts energy from one form to another, typically converting a non-electrical physical parameter (e.g., temperature, pressure, displacement) into an electrical signal (e.g., voltage, current, resistance).

Feature / Criteria	Active Transducers	Passive Transducers
Power Requirement	Self-generating; do not require any external electrical source.	Externally powered; require an auxiliary AC/DC source.
Operating Principle	Convert physical energy directly to electrical energy.	Cause a change in electrical parameters (R, L, or C).
Signal Source	Draw energy from the process under measurement.	Draw energy from the external auxiliary source.
Output Type	Analog voltage or current signals (e.g., mV, pA).	Parameter variation (e.g., ΔR , ΔL , ΔC).
Examples	Thermocouple, Piezoelectric crystal, Solar cell.	RTD, LVDT, Strain gauge, Thermistor.

1.1.2 2. Linear Variable Differential Transformer (LVDT)

The **Linear Variable Differential Transformer (LVDT)** is a passive inductive transducer used to measure linear displacement with high resolution and accuracy.

A. Construction & Operating Principle

An LVDT consists of a single primary winding (P) and two secondary windings (S_1 and S_2) wound symmetrically on a hollow cylindrical coil form. A highly permeable ferromagnetic core is placed inside the cylinder and attached to the moving element whose displacement is to be measured.

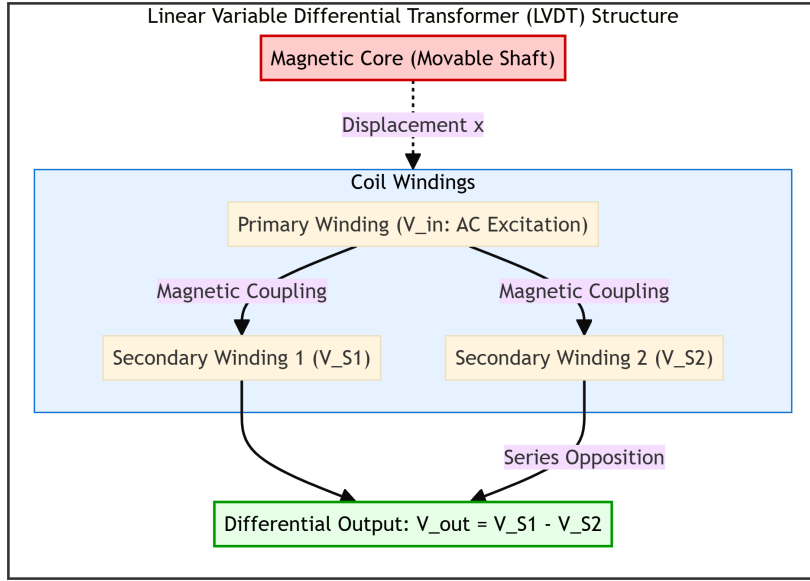


Figure 1.1: LVDT Electro-Mechanical Schematic

The primary winding is excited by an alternating current (AC) voltage source:

$$V_{in} = V_p \sin(\omega t)$$

This excitation creates an alternating magnetic field that induces voltages in both secondary windings through mutual inductance. The two secondary windings are connected in **series opposition**, meaning their output voltages oppose each other. The net differential output voltage (V_{out}) is:

$$V_{out} = V_{S1} - V_{S2}$$

B. Electro-Mechanical Transfer Characteristics

The position of the core determines the mutual inductance between the primary and each secondary winding:

1. **Null Position (Core at Center):** The core is symmetrically placed. The magnetic coupling to both secondary windings is identical, meaning $V_{S1} = V_{S2}$. Thus, the differential output voltage is:

$$V_{out} = 0 \text{ V}$$

(Note: In practice, a small residual null voltage exists due to stray capacitances and harmonic frequencies).

1. **Left Displacement (-x):** The core shifts toward S_1 . The magnetic coupling to S_1 increases while coupling to S_2 decreases, resulting in $V_{S1} > V_{S2}$. The output voltage is in-phase with the primary voltage.
2. **Right Displacement (+x):** The core shifts toward S_2 . The magnetic coupling to S_2 increases while coupling to S_1 decreases, resulting in $V_{S2} > V_{S1}$. The output voltage is 180° out-of-phase with the primary voltage.

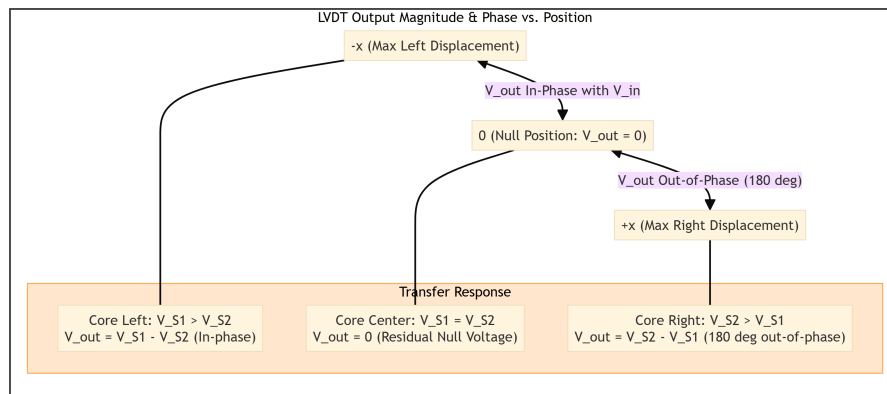


Figure 1.2: LVDT Transfer Characteristics

C. Advantages & Disadvantages

- **Advantages:**
- **Infinite Resolution:** Linear movement produces a continuous, infinite-resolution output signal.
- **Low Friction & Wear:** No physical electrical contact between the core and the coil assembly.
- **Ruggedness:** Capable of surviving extreme shock, vibration, and environmental contamination.
- **Low Hysteresis:** Highly repeatable displacement measurements.
- **Disadvantages:**
- **Magnetic Interference:** Vulnerable to external stray magnetic fields; requires magnetic shielding.
- **Temperature Sensitivity:** Changes in temperature alter the coil resistance and magnetic properties of the core.
- **AC Source Required:** Demodulator circuitry is needed to convert the AC output to a DC signal.

1.1.3 3. Temperature Sensors

Industrial temperature measurement uses different transducers depending on the temperature range, sensitivity, and linearity required.

A. Thermocouples

A **thermocouple** is an active temperature transducer based on the **Seebeck Effect**: when two dissimilar metal conductors are joined at two junctions held at different temperatures (T_{hot} and T_{cold}), an electromotive force (EMF) is generated that is proportional to the temperature difference:

$$E = a(T_{hot} - T_{cold}) + b(T_{hot} - T_{cold})^2$$

where a and b are constants related to the thermocouple metals.

Common Thermocouple Types:

1. Type K (Chromel / Alume1):

- *Composition:* Nickel-Chromium (+) and Nickel-Aluminum (-).
- *Temperature Range:* -200°C to $+1372^{\circ}\text{C}$.
- *Application:* Widely used due to its oxidation resistance at high temperatures.

1. Type J (Iron / Constantan):

- *Composition:* Iron (+) and Copper-Nickel (-).
- *Temperature Range:* -40°C to $+750^{\circ}\text{C}$.
- *Application:* Suitable for vacuum or reducing atmospheres.

Cold Junction Compensation (CJC): Thermocouple tables are calibrated with the reference (cold) junction at 0°C . In industrial applications, the cold junction is at the ambient temperature of the control panel (T_c). To measure the absolute hot junction temperature (T_h), **Cold Junction Compensation** is applied:

1. The ambient temperature (T_c) is measured using a local sensor (e.g., RTD or thermistor).
2. A compensating voltage V_{comp} corresponding to the thermocouple's output at T_c relative to 0°C is calculated.
3. This voltage is added to the raw thermocouple voltage (V_{raw}) to determine the true temperature:

$$V_{net} = V_{raw} + V_{comp}$$

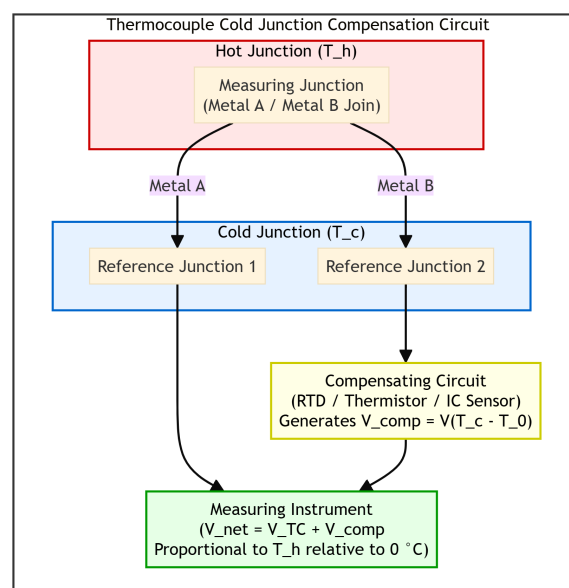


Figure 1.3: Thermocouple & Cold Junction Compensation

B. RTD (Resistance Temperature Detector)

An **RTD** is a passive sensor whose resistance increases linearly with temperature (Positive Temperature Coefficient, PTC). The relationship is governed by:

$$R_T = R_0[1 + \alpha(T - T_0)]$$

where R_0 is the resistance at reference temperature T_0 , and α is the temperature coefficient of resistance.

- *Material:* Platinum (PT100 has a nominal resistance of 100 Ω at 0°C).
- *Pros & Cons:* Highly stable, accurate, and linear; but has slow response time and is subject to self-heating errors.

C. Thermistors

A **thermistor** is a semiconductor temperature sensor that exhibits a highly non-linear resistance change with temperature, typically possessing a Negative Temperature Coefficient (NTC).

- *Formula (Steinhart-Hart Equation):*

$$\frac{1}{T} = A + B \ln(R) + C(\ln(R))^3$$

- *Pros & Cons:* Extremely sensitive and fast-acting; but highly non-linear with a limited temperature range.

D. Pressure Thermometer

A **pressure thermometer** is a filled-system thermometer that measures temperature by detecting the thermal expansion of a working fluid.

- *Operation:* A bulb containing a liquid, gas, or vapor is placed in the thermal zone. As temperature rises, the fluid expands, creating a pressure increase. This pressure is transmitted via a capillary tube to an elastic pressure sensing element (such as a Bourdon tube or bellows), which mechanically moves a pointer or electronic transmitter.

1.1.4 4. Pressure Sensors

Industrial pressure measurement relies on elastic sensing elements that deform under applied pressure, transferring this mechanical movement to electrical transducers.

1. **Diaphragm:** A thin, flexible circular plate that deflects under pressure. Diaphragms are used for low-pressure measurements and are often coupled with strain gauges or capacitive sensors to convert deflection into electrical signals.
2. **Bellows:** Thin-walled metallic cylinders with deep convolutions. When pressure is applied internally, the bellows expands axially, acting as a spring. It provides a larger mechanical displacement than a diaphragm and is ideal for medium pressure ranges.
3. **Bourdon Tube:** A C-shaped, helical, or spiral tube of oval cross-section. When pressurized internally, the tube tends to straighten out. The free end of the tube moves in proportion to the applied pressure, which is mechanically linked to a pointer or electronic potentiometer.

1.1.5 5. Force Sensors & Strain Gauges

A **strain gauge** is a passive resistive sensor used to measure mechanical strain (deformation).

A. Working Principle & Gauge Factor Derivation

When a metallic wire is subjected to mechanical tension, its length (L) increases, its cross-sectional area (A) decreases, and its resistivity (ρ) changes due to piezoresistive effects. The resistance of the wire is:

$$R = \frac{\rho L}{A}$$

Taking the natural logarithm and differentiating yields:

$$\frac{dR}{R} = \frac{dL}{L} - \frac{dA}{A} + \frac{d\rho}{\rho}$$

Since the cross-sectional area of a wire with diameter D is $A = \pi D^2/4$, we have $\frac{dA}{A} = 2\frac{dD}{D}$. Substituting this gives:

$$\frac{dR}{R} = \frac{dL}{L} - 2\frac{dD}{D} + \frac{d\rho}{\rho}$$

Poisson's ratio (ν) is defined as the ratio of lateral strain to longitudinal strain:

$$\nu = -\frac{dD/D}{dL/L} \implies \frac{dD}{D} = -\nu \frac{dL}{L}$$

Substituting Poisson's ratio back into the equation:

$$\frac{dR}{R} = \frac{dL}{L}(1 + 2\nu) + \frac{d\rho}{\rho}$$

Dividing the entire equation by the longitudinal strain ($\epsilon = dL/L$) yields the **Gauge Factor (G)**, which represents the sensitivity of the strain gauge:

$$G = \frac{dR/R}{dL/L} = 1 + 2\nu + \frac{d\rho/\rho}{dL/L}$$

where:

- 1: Change in resistance due to change in length.
- 2ν : Change in resistance due to change in area.
- $\frac{d\rho/\rho}{dL/L}$: Change in resistance due to the piezoresistive effect.

For metal wire strain gauges, G is typically around 2.0. For semiconductor strain gauges, the piezoresistive component is very large, yielding a gauge factor of 100–150.

1.1.6 6. Ultrasonic Sensors

An **ultrasonic sensor** is a non-contact sensor used for distance measurement and object detection.

A. Working Principle

The sensor consists of a piezoelectric transmitter that emits high-frequency sound wave pulses (typically in the **10–65 kHz** range) and a receiver that waits for the reflected echo from the target object.

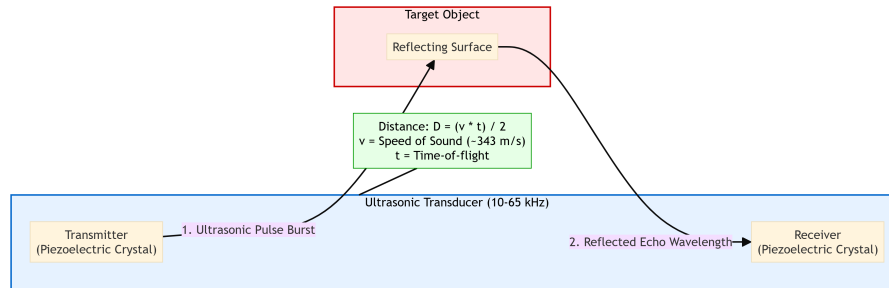


Figure 1.4: Ultrasonic Sensor Working Principle

The distance (D) to the object is calculated based on the **Time-of-Flight (ToF)** of the sound wave:

$$D = \frac{v \times t}{2}$$

where:

- v : Speed of sound in air (approximately 343 m/s at 20°C).
- t : Measured time delay between the transmission of the pulse and the reception of the echo.

B. Applications

- **Liquid Level Measurement:** Monitoring level in chemical tanks without physical contact, preventing corrosion issues.
- **Obstacle Avoidance:** Used in industrial automated guided vehicles (AGVs) and robotics.
- **Non-Destructive Testing (NDT):** Detecting internal cracks and voids in metallic structures.

1.2 Section 2: Industrial Actuators [15M] [Priority: High]

An **actuator** is a device that converts an electrical, pneumatic, or hydraulic control signal into mechanical energy (motion) to manipulate a process.

1.2.1 1. DC Motors & Starters

A **DC motor** converts electrical direct current into mechanical rotation.

A. Back EMF and High Starting Current

When a DC motor rotates, its armature conductors cut the magnetic field lines, inducing an internal voltage called **Back EMF** (E_b) that opposes the applied terminal voltage (V):

$$E_b = \frac{\Phi Z N P}{60 A} = K_m \omega$$

where N is motor speed.

The armature current (I_a) is governed by:

$$I_a = \frac{V - E_b}{R_a}$$

where R_a is the armature winding resistance (typically very small, $< 1 \Omega$).

At start-up, the motor speed is zero ($N = 0$), which means the Back EMF is also zero ($E_b = 0$). The starting current (I_{start}) is:

$$I_{start} = \frac{V}{R_a}$$

Because R_a is extremely small, this starting current can be 5–10 times larger than the rated full-load current. This high current can:

- Burn out the armature windings due to excessive $I^2 R_a$ heating.
- Cause severe sparking at the commutator and damage the brushes.
- Produce a voltage drop on the power supply line, affecting other connected equipment.

B. Need for Motor Starters

To prevent armature damage, a **motor starter** (such as a 3-point or 4-point starter) is connected in series with the armature. The starter inserts a variable resistance (R_{start}) into the armature circuit at boot to limit the starting current:

$$I_{start} = \frac{V}{R_a + R_{start}}$$

As the motor accelerates, the Back EMF (E_b) builds up. The starter mechanism gradually cuts out the starting resistors until the armature is connected directly across the supply line once the motor reaches operating speed.

1.2.2 2. Stepper Motors vs. Servo Motors

Both stepper and servo motors are used for precise position control in automation, but their operating principles and application areas differ significantly.

Feature / Criteria	Stepper Motor	Servo Motor
Control Loop	Open-loop (no feedback sensor needed).	Closed-loop (requires encoder/resolver feedback).
Positioning Method	Moves in discrete, defined steps.	Continuous rotation guided by feedback error signals.
Torque Profile	High holding torque at standstill; falls off at high speeds.	Consistent torque output across the entire speed range.
Overshooting	No overshooting; locks into step position.	May overshoot slightly; relies on PID tuning to settle.
Cost & Complexity	Simple drive electronics; low cost.	Complex controller, encoder, and tuning; high cost.

A. Stepper Motor Step Modes

A stepper motor rotates by sequentially energizing electromagnetic stator phases. The step resolution can be modified using different excitation modes:

1. **Full-Step (2-Phase ON):** Stator phases are energized in pairs. The motor moves through its basic step angle (e.g., 1.8° per step), providing maximum torque.
2. **Half-Step:** Alternates between energizing one phase and two phases (e.g., $A \rightarrow AB \rightarrow B \rightarrow BC$). This halves the step angle (e.g., 0.9°), doubling the positioning resolution and reducing vibrations.
3. **Micro-stepping:** The current supplied to the windings is modulated as sine and cosine waves. This allows the rotor to position itself between physical steps, yielding extremely high resolution, quiet operation, and smooth rotation at low speeds.



Figure 1.5: Stepper Motor Step Modes

1.2.3 3. Piezoelectric Actuators

A **piezoelectric actuator** converts electrical energy directly into linear displacement using the **Inverse Piezoelectric Effect**: when an electric field is applied across a piezoelectric crystal (e.g., Lead Zirconate Titanate - PZT), the crystal lattice deforms, producing a mechanical displacement.

A. Types of Piezoelectric Actuators

- **Stack Actuators:** Consist of multiple piezoelectric disks stacked on top of each other in series mechanically and in parallel electrically. They produce high pushing forces (up to tens of kilonewtons) but have very short displacements (typically $< 100 \mu\text{m}$).
- **Stripe (Bending) Actuators:** Consist of two thin piezoelectric strips bonded together. When voltage is applied, one strip expands while the other contracts, causing the assembly to bend. They produce larger displacements (millimeters) but much lower forces.

B. Pros & Cons

- **Advantages:** Sub-nanometer positioning resolution, fast response times (microseconds), and high force output.
- **Disadvantages:** Very short stroke length, significant hysteresis (10–15%), and requires high driving voltages (> 100 V).

1.2.4 4. Pneumatic Actuators

A **pneumatic actuator** uses compressed air to produce mechanical motion.

A. Spring-Diaphragm Actuator Working Principle

The most common control valve actuator is the spring-diaphragm actuator. It consists of a flexible diaphragm held within a metal casing, backed by a heavy range spring.

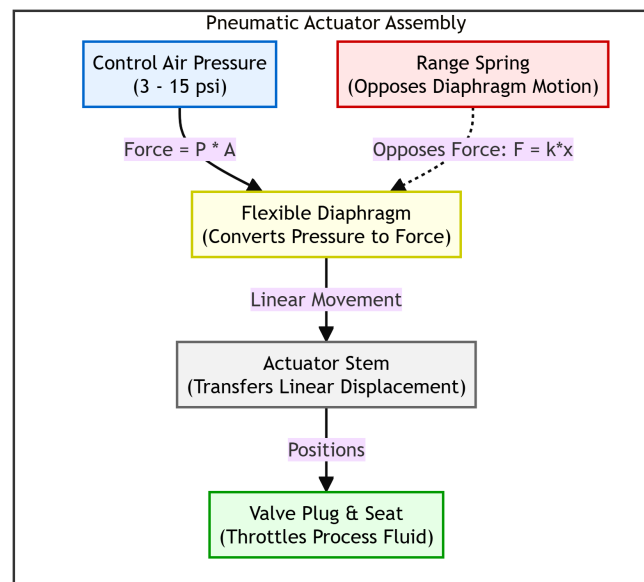


Figure 1.6: Pneumatic Spring-Diaphragm Actuator

- **Operation:** Compressed air (typically an industrial standard **3–15 psi** signal) is introduced into the chamber on one side of the diaphragm. The pressure exerts a force ($F = P \times A_{diaphragm}$) that compresses the range spring. The actuator stem moves linearly in proportion to the air pressure.
- **Fail-Safe Action:**
- **Air-to-Open (Fail-Closed):** Increasing pressure moves the stem downward to open the valve; loss of air pressure causes the spring to push the stem up, closing the valve.
- **Air-to-Close (Fail-Open):** Increasing pressure closes the valve; loss of air pressure causes the spring to push the stem up, opening the valve.

1.3 Section 3: Signal Conditioning & Transmission [15M] [Priority: High]

1.3.1 1. Signal Conditioning System

Signal conditioning refers to the operations performed on raw sensor outputs to make them suitable for transmission, display, or analog-to-digital conversion.

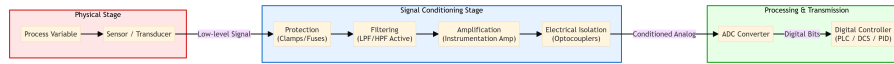


Figure 1.7: Signal Conditioning Functional Architecture

Core Functions of the Block Diagram:

1. **Protection:** Guards downstream electronics against high-voltage transients or current spikes (e.g., using zener diodes, varistors, or fuses).
2. **Filtering:** Removes unwanted high-frequency noise or power line interference (50 Hz) and prevents aliasing before digitization.
3. **Amplification:** Boosts low-voltage sensor outputs (such as mV-level thermocouple signals) to a standard range (e.g., 0–5 V or 0–10 V) using high-performance instrumentation amplifiers.
4. **Isolation:** Breaks ground loops and protects the controller from high common-mode voltages by separating the sensor circuit from the controller circuit galvanically (using optocouplers or isolation transformers).
5. **Analog-to-Digital Conversion (ADC):** Converts the conditioned continuous analog voltage into a discrete digital representation for processing by microprocessors, PLCs, or DCS controllers.

1.3.2 2. Active Filter Design using OP-AMPs

Filters are used in signal conditioning to isolate specific frequency bands and suppress electrical noise. **Active filters** use operational amplifiers (OP-AMPs) combined with resistor-capacitor (RC) networks, offering high input impedance, low output impedance, and signal gain.

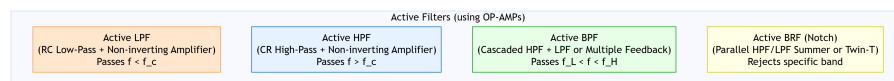


Figure 1.8: Active Filter Circuit Topologies

A. Active Low-Pass Filter (LPF)

An active LPF passes low-frequency signals and attenuates frequencies above a specific cut-off frequency (f_c).

- **Circuit:** A passive RC low-pass filter connected to the non-inverting input of an OP-AMP configured as a voltage amplifier.

- **Transfer Function:**

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{A_v}{1 + sRC}$$

where $A_v = 1 + \frac{R_f}{R_1}$ is the passband voltage gain, and $s = j\omega$.

- **Cut-off Frequency:**

$$f_c = \frac{1}{2\pi RC}$$

B. Active High-Pass Filter (HPF)

An active HPF passes high-frequency signals and blocks low-frequency signals below its cut-off frequency.

- **Circuit:** Created by swapping the resistor and capacitor in the LPF input network.
- **Transfer Function:**

$$H(s) = \frac{A_v sRC}{1 + sRC}$$

- **Cut-off Frequency:**

$$f_c = \frac{1}{2\pi RC}$$

C. Active Band-Pass Filter (BPF)

A BPF allows a specific range of frequencies to pass while attenuating all others.

- **Design:** Constructed by cascading an active HPF with a lower cut-off frequency (f_{c1}) followed by an active LPF with a higher cut-off frequency (f_{c2}):

$$f_{c1} < f < f_{c2}$$

D. Active Band-Reject (Notch) Filter (BRF)

A BRF attenuates a specific narrow band of frequencies (e.g., 50 Hz power line hum) while passing all other frequencies.

- **Design:** Designed by summing the outputs of an active LPF and an active HPF in parallel using a summing amplifier, or by implementing a Twin-T notch filter topology.

1.3.3 3. Sensor Protection Circuits

Protection circuits prevent industrial voltage transients, short circuits, and electrostatic discharges (ESD) from damaging expensive controller hardware.

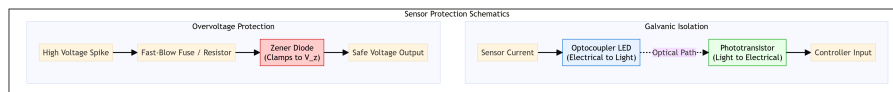


Figure 1.9: Sensor Protection Circuits

- **Overvoltage Protection (Zener Clamp):** A Zener diode with breakdown voltage V_z is placed in parallel with the sensor output line. If a voltage spike exceeds V_z , the Zener diode breaks down and shunts the excess current to ground, clamping the output voltage to a safe level. A fast-blow fuse or series resistor limits the maximum current.
- **Galvanic Isolation (Optocoupler):** Converts the electrical sensor signal into light using an LED, which is transmitted across an air gap to a phototransistor. Because there is no physical electrical connection between the input and output sides, voltage surges of up to several kilovolts on the sensor side cannot pass to the controller.

1.3.4 4. Noise Suppression & Signal Transmission

Industrial plants are filled with electromagnetic interference (EMI) from heavy machinery, motors, and switching circuits.

- **Twisted-Pair Cabling:** Twisting the signal and return wires together ensures that any induced electromagnetic noise affects both wires equally (common-mode noise). This noise can then be rejected downstream using differential amplifiers (high Common-Mode Rejection Ratio - CMRR).
- **Shielding:** Wrapping signal cables in a conductive metal foil or braided sheath grounded at a single point diverts electrostatic noise currents to ground.
- **4-20 mA Current Loop Transmission:** Sending signals as a current rather than a voltage makes the transmission immune to line resistance and voltage drops over long distances. A line break is easily detected because the minimum signal level is 4 mA rather than 0 mA.

1.4 Section 4: Estimation of Errors & Calibration [5M] [Priority: Medium]

1.4.1 1. Calibration

Calibration is the process of comparing the output of a measurement instrument against a known, traceable reference standard of higher accuracy under specified environmental conditions.

- **Importance:** Ensures measurement accuracy, establishes traceability to national standards, detects drift over time, and maintains safety compliance in process loops.

1.4.2 2. Error Classification

Measurement errors represent the difference between the measured value and the true value of a variable.

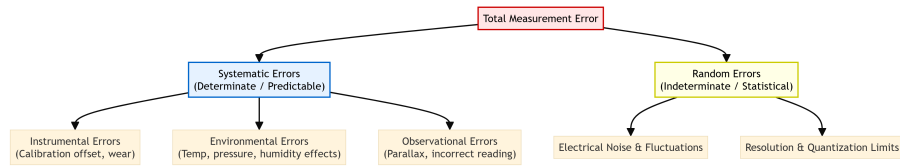


Figure 1.10: Measurement Error Taxonomy

1. **Systematic (Determinate) Errors:** Consistent, predictable deviations that occur in one direction. They can be identified and corrected through calibration or compensation.
 - *Instrumental:* Zero offset, worn components, or bad calibration.
 - *Environmental:* Changes in ambient temperature, pressure, or humidity that affect the sensor.
 - *Observational:* Parallax errors when reading analog meters.
1. **Random (Indeterminate) Errors:** Unpredictable, fluctuating deviations caused by random noise or minor environmental variations. They cannot be eliminated but can be analyzed and reduced statistically by taking multiple measurements and calculating the mean.

1.4.3 3. Limiting Error (Uncertainty Propagation)

The **limiting error** (or guarantee error) is the maximum deviation from the nominal value specified by the manufacturer. When multiple measured parameters with individual limiting errors are combined to calculate a system result, the total limiting error propagates.

Let x_1 and x_2 be two independent variables with limiting errors Δx_1 and Δx_2 (relative limiting errors $\delta_1 = \Delta x_1/x_1$ and $\delta_2 = \Delta x_2/x_2$).

A. Summation or Subtraction

Let the calculated result be:

$$X = x_1 \pm x_2$$

Differentiating the equation:

$$dX = dx_1 \pm dx_2$$

To find the worst-case maximum limiting error (ΔX), we sum the absolute values of the individual errors:

$$\Delta X = \Delta x_1 + \Delta x_2$$

The absolute limiting error of a sum or difference is the **sum of the absolute limiting errors** of the individual components.

B. Multiplication or Division

Let the calculated result be:

$$X = x_1 \cdot x_2 \quad \text{or} \quad X = \frac{x_1}{x_2}$$

Taking the natural logarithm and differentiating:

$$\ln(X) = \ln(x_1) \pm \ln(x_2) \implies \frac{dX}{X} = \frac{dx_1}{x_1} \pm \frac{dx_2}{x_2}$$

To find the worst-case maximum relative limiting error, we sum the absolute values of the relative errors:

$$\frac{\Delta X}{X} = \frac{\Delta x_1}{x_1} + \frac{\Delta x_2}{x_2}$$

The relative limiting error of a product or quotient is the **sum of the relative limiting errors** of the individual components.

1.5 Section 7: Standard Process Signal Transmission & Converters (Q1.9) [5M][[]]

In industrial process control, measurement signals from sensors are converted to standard transmission ranges to ensure compatibility and reliability across long distances.

1.5.1 1. The 4–20 mA Current Loop Standard

The **4–20 mA DC current loop** is the dominant analog signal transmission standard in modern process industries.

- **Operating Range:** A process variable's minimum value corresponds to 4 mA (active zero), and its maximum value corresponds to 20 mA.
- **Why 4 mA Active Zero?**
 1. *Fault Detection:* If a wire breaks or a sensor fails completely, the loop current drops to 0 mA, immediately triggering a "line break" alarm. A system using 0–20 mA cannot distinguish between a minimum process reading and a severed connection.
 2. *Self-Powered Transmitters:* The 4 mA standby current is sufficient to power the internal circuitry of two-wire field transmitters without needing separate power lines.
- **Core Components:**
 1. **DC Power Supply:** Provides the driving voltage (typically 24 V DC) to overcome line resistances.
 2. **Two-Wire Transmission Path:** Copper wires carrying the loop current.
 3. **Transmitter:** Regulates the loop current between 4 and 20 mA in proportion to the sensor's reading.
 4. **Receiver / Load Resistor:** The controller input card (e.g., 250 Ω load resistor converting the 4–20 mA current back to a standard 1–5 V voltage for the ADC).

1.5.2 2. The 3–15 psi Pneumatic Standard

- **Operating Range:** Primarily used in historic systems or safety-critical valves. The process signal is carried by compressed air between **3 and 15 psi** (pounds per square inch).
- **Why 3 psi Active Zero?** Similar to the current loop, 3 psi is used as an active zero to detect pneumatic line leaks. If air pressure falls to 0 psi, a line failure is registered.

1.5.3 3. Converters (I/P and P/I)

Because modern controllers are electronic but many large control valves remain pneumatic, converters are essential:

- **Current-to-Pneumatic (I/P) Converter:** Receives an electronic 4–20 mA command signal from the controller and outputs a proportional 3–15 psi compressed air signal to drive a pneumatic actuator.
- **Pneumatic-to-Current (P/I) Converter:** Receives a 3–15 psi pressure measurement and converts it to a standard 4–20 mA electrical current signal for transmission to a centralized controller.

1.6 Section 8: Capacitive & Potentiometric Displacement Sensors (Q1.10) [5M][[]]

For precise physical position tracking, capacitive and potentiometric transducers offer simple and highly sensitive solutions.

1.6.1 1. Capacitive Displacement Sensors

A capacitive sensor consists of two parallel metal plates separated by a dielectric material. The capacitance C is governed by:

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

Where:

- ϵ_0 : Permittivity of free space.
- ϵ_r : Relative permittivity of the dielectric.
- A : Overlapping area of the plates.
- d : Separation distance between the plates.

Displacement can be measured by varying one of three parameters:

1. **Variable Separation (d):** One plate remains stationary while the other moves axially in response to displacement. The capacitance is inversely proportional to d . This method is extremely sensitive, used to measure sub-micron displacements.
2. **Variable Overlapping Area (A):** One plate slides laterally relative to the other, varying the overlapping area. The capacitance changes linearly with lateral displacement. Ideal for larger linear and angular displacements.

3. **Variable Dielectric Permittivity (ϵ_r):** A slab of dielectric material is moved between two fixed plates. Since the dielectric constant of the slab is different from air, its position alters the net capacitance. Commonly used for liquid level sensors.

1.6.2 2. Potentiometric Sensors

A **potentiometer** is a passive displacement transducer consisting of a resistive element with a sliding contact (wiper) mechanically linked to the moving body.

- **Linear Potentiometer:** Measures linear movements. The resistance between the wiper and one end terminal changes linearly with axial wiper displacement.
- **Angular (Rotary) Potentiometer:** Measures rotational angles. The resistive path is circular, and the wiper tracks angular changes.
- **Working Principle:** Configured as a voltage divider. A stable reference voltage V_{ref} is applied across the outer terminals, and the output voltage V_{out} is read from the wiper. It varies linearly with position:

$$V_{out} = V_{ref} \left(\frac{x}{L} \right)$$

where x is displacement and L is total potentiometer length.

Chapter 2

Controller Tuning & Implementation

2.1 Section 1: Process Control Operations & Modes [5M] [Priority: Medium]

2.1.1 1. Servo vs. Regulatory Operations

Process control loops are designed to maintain a controlled variable at a desired value (setpoint) despite environmental variations. Depending on the source of the excitation, process control operations are divided into two main categories:

Feature / Criteria	Servo Operation (Setpoint Tracking)	Regulatory Operation (Disturbance Rejection)
Core Objective	Keep the process variable equal to a dynamically changing setpoint.	Keep the process variable constant at a fixed setpoint despite load disturbances.
Primary Input Change	Setpoint ($R(s)$) changes; Load Disturbance ($D(s)$) remains constant.	Load Disturbance ($D(s)$) changes; Setpoint ($R(s)$) remains constant.
Industrial Example	Changing the temperature setpoint of a furnace during a heat treatment cycle.	Maintaining a constant product temperature in a heat exchanger despite fluctuations in inlet flow.
Response Characteristic	Fast tracking of the setpoint path with minimal overshoot and lag.	Fast return to the steady-state setpoint after a disturbance deflection.

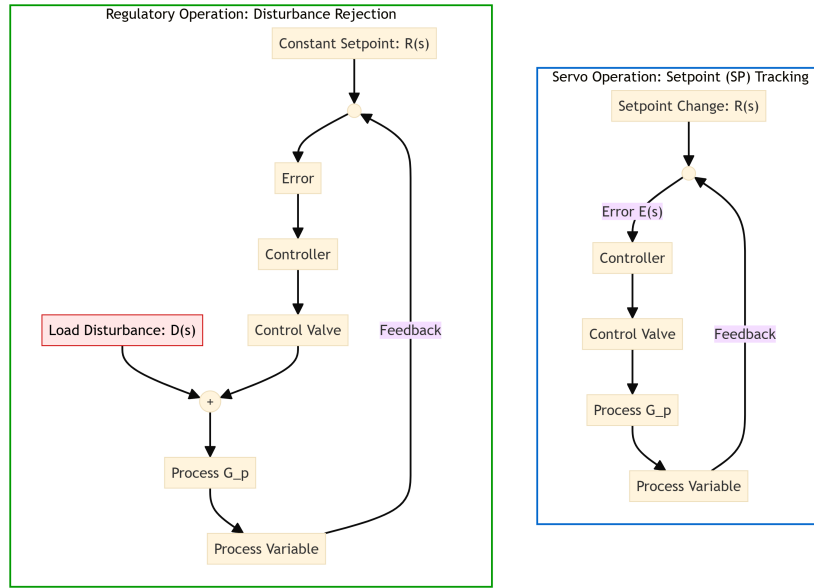


Figure 2.1: Servo vs. Regulatory Control Loops

2.1.2 2. Self-Regulating vs. Non-Self-Regulating Systems

- **Self-Regulating Systems:** A process that naturally finds a new steady-state equilibrium following a step change in input without any controller intervention.
- *Example:* A liquid storage tank with gravity-driven outflow. If the inlet flow rate (F_{in}) is stepped up, the liquid level (h) rises. As the level rises, the hydrostatic pressure increases, which drives a faster outflow rate (F_{out}). The level continues to rise until $F_{out} = F_{in}$, reaching a new stable level equilibrium.
- **Non-Self-Regulating (Integrating) Systems:** A process that does not reach a new equilibrium after a step input change but instead continuously ramps or drifts until it hits a physical limit (saturation).
- *Example:* A liquid storage tank with a constant displacement pump at the outflow. If the inlet flow (F_{in}) is stepped up to exceed the fixed outflow pump capacity, the liquid level will continuously ramp upward until the tank overflows.

2.1.3 3. Process Dynamics: Dead-Time Element

Dead-time (or transport lag) is the time delay between the application of an input signal and the first observable change in the process variable. It is common in industrial systems involving material transport (e.g., fluid flow through pipes, conveyor belts).

- **Laplace Transfer Function:**

$$G(s) = e^{-sT_d}$$

where T_d is the dead-time delay.

- **Padé Approximation:**

Because the exponential term e^{-sT_d} is non-rational, it is difficult to analyze in standard transfer function calculations. A first-order Padé approximation is commonly used:

$$e^{-sT_d} \approx \frac{1 - \frac{T_d}{2}s}{1 + \frac{T_d}{2}s}$$

2.2 Section 2: Controller Modes [15M] [Priority: High]

A controller detects the error between the Setpoint (SP) and the Process Variable (PV) and calculates a control output to adjust the final control element (valve).

$$e(t) = SP(t) - PV(t)$$

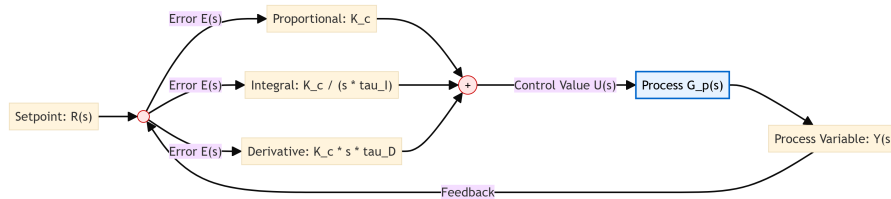


Figure 2.2: Parallel PID Control Loop Block Diagram

2.2.1 1. Proportional (P) Controller

The controller output ($p(t)$) is directly proportional to the error signal ($e(t)$).

- **Equation:**

$$p(t) = K_c e(t) + p_0$$

where K_c is the Proportional Gain, and p_0 is the bias (controller output when error is zero).

- **Proportional Band (PB):**

The range of error input that causes the controller output to shift from 0% to 100%.

$$PB = \frac{100}{K_c} \%$$

- **Characteristics:**

- A very wide PB (low K_c) results in a slow, sluggish response.
- A narrow PB (high K_c) makes the system faster but increases oscillation.
- **Steady-State Offset:** A proportional controller always exhibits a steady-state error (offset) following a load change because a non-zero error is required to sustain a controller output other than p_0 .

2.2.2 2. Integral (I) Controller

The controller output is proportional to the accumulation of the error signal over time.

- **Equation:**

$$p(t) = \frac{1}{\tau_I} \int_0^t e(t') dt' + p_0$$

where τ_I is the integral (reset) time. The reset rate is defined as $1/\tau_I$ (repeats per minute).

- **Characteristics:**

- **Offset Elimination:** Since the controller output continues to change as long as any error exists, the integral mode completely eliminates steady-state offset.
- **Sluggishness:** The integral action introduces a -90° phase lag, which slows down the transient response and can destabilize the loop.
- **Reset Windup:** If a large error persists for a long time (e.g., during start-up or valve saturation), the integrator continues to accumulate error, driving the controller output to its limit. The system will experience heavy overshoot as the controller takes time to "unwind" the accumulated value.

2.2.3 3. Derivative (D) Controller

The controller output is proportional to the rate of change of the error signal.

- **Equation:**

$$p(t) = \tau_D \frac{de(t)}{dt}$$

where τ_D is the derivative (rate) time.

- **Characteristics:**

- **Anticipatory Action:** By looking at the slope of the error, the derivative mode predicts where the process is heading and applies early corrective action, reducing overshoot and settling time.
- **Steady-State Limitation:** If the error is constant (non-zero but flat), the derivative output is zero. Thus, it cannot eliminate steady-state offset on its own.
- **Noise Amplification:** High-frequency measurement noise (small amplitude but very high rate of change) is heavily amplified by the derivative term, causing control valve wear.

2.2.4 4. Composite Controller Modes (PI, PD, PID)

To combine the speed of proportional, the offset elimination of integral, and the stabilization of derivative, composite controller modes are used:

- **Proportional-Integral (PI) Controller:**

- *Equation:*

$$p(t) = K_c \left[e(t) + \frac{1}{\tau_I} \int_0^t e(t') dt' \right] + p_0$$

- *Characteristics:* Eliminates steady-state offset and improves steady-state accuracy, but introduces a phase lag that slows down the transient settling time.

- **Proportional-Derivative (PD) Controller:**

- *Equation:*

$$p(t) = K_c \left[e(t) + \tau_D \frac{de(t)}{dt} \right] + p_0$$

- *Characteristics:* Increases system damping, reduces overshoot, and accelerates the transient response, but cannot eliminate steady-state offset.

- **Proportional-Integral-Derivative (PID) Controller:**

- *Equation:*

$$p(t) = K_c \left[e(t) + \frac{1}{\tau_I} \int_0^t e(t') dt' + \tau_D \frac{de(t)}{dt} \right] + p_0$$

- *Transfer Function (Parallel Form):*

$$G_c(s) = K_c \left[1 + \frac{1}{\tau_I s} + \tau_D s \right]$$

- *Characteristics:* Provides the optimal balance of offset elimination, speed, stability, and disturbance rejection.

2.3 Section 3: Analog & Digital Controller Implementation [15M] [Priority: High]

Controllers are implemented physically using electronic OP-AMP circuits, pneumatic flapper-bellows systems, or digital microcontrollers.

2.3.1 1. Electronic OP-AMP Implementation

Active operational amplifiers (OP-AMPs) can implement P, I, D, and composite controllers by configuring resistors and capacitors in the input and feedback paths.

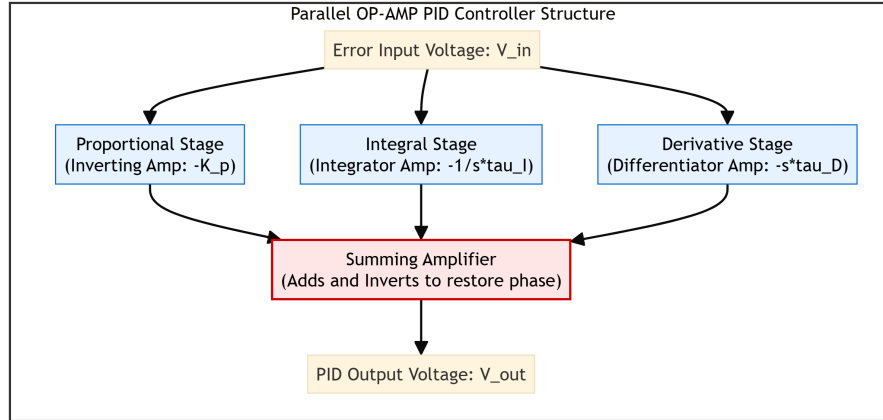


Figure 2.3: Parallel OP-AMP PID Controller

A. Proportional (P) Controller Circuit

- **Schematic:** An inverting amplifier configuration.
- Input path: Resistor R_{in} connected to the inverting terminal.
- Feedback path: Resistor R_f connected from the output to the inverting terminal.
- Non-inverting terminal is grounded.
- **Transfer Function:**

$$\frac{V_{out}(s)}{V_{in}(s)} = -\frac{R_f}{R_{in}}$$

- **Proportional Gain:**

$$K_p = \frac{R_f}{R_{in}}$$

B. Integral (I) Controller Circuit

- **Schematic:** An integrating amplifier configuration.
- Input path: Resistor R .
- Feedback path: Capacitor C .
- **Transfer Function:**

$$\frac{V_{out}(s)}{V_{in}(s)} = -\frac{1}{RCs}$$

- **Integration Time Constant (τ_I):**

$$\tau_I = RC$$

- **Parameter Analysis:**

If the resistor (R) and capacitor (C) values are both doubled ($2R$ and $2C$):

- The new integration time constant is:

$$\tau_I' = (2R)(2C) = 4RC = 4\tau_I$$

The integration time constant is **quadrupled**.

- Since the integrator gain is $K_I = 1/\tau_I$, the controller gain is **quartered** (halved twice), meaning the system's integration rate slows down significantly.

C. Derivative (D) Controller Circuit

- **Schematic:** A differentiating amplifier configuration.
- Input path: Capacitor C .
- Feedback path: Resistor R .
- **Transfer Function:**

$$\frac{V_{out}(s)}{V_{in}(s)} = -RCs$$

- **Derivative Time Constant (τ_D):**

$$\tau_D = RC$$

D. Proportional-Integral (PI) Controller Circuit

- **Schematic:** An inverting amplifier with a resistor R_1 in the input path and a series resistor-capacitor network (R_f and C_f) in the feedback path.
- **Transfer Function:**

$$\frac{V_{out}(s)}{V_{in}(s)} = -\frac{Z_f}{Z_{in}} = -\frac{R_f + \frac{1}{sC_f}}{R_1} = -\frac{R_f}{R_1} \left[1 + \frac{1}{R_f C_f s} \right]$$

- **Parameters:**

$$K_p = \frac{R_f}{R_1}, \quad \tau_I = R_f C_f$$

E. Proportional-Derivative (PD) Controller Circuit

- **Schematic:** Input path has a parallel resistor-capacitor network (R_1 and C_1), and the feedback path has a resistor R_f .
- **Transfer Function:**

$$\frac{V_{out}(s)}{V_{in}(s)} = -\frac{R_f}{Z_{in}} = -R_f \left[\frac{1}{R_1} + sC_1 \right] = -\frac{R_f}{R_1} [1 + R_1 C_1 s]$$

- **Parameters:**

$$K_p = \frac{R_f}{R_1}, \quad \tau_D = R_1 C_1$$

2.3.2 2. Pneumatic PID Controller Implementation

In oil refineries and chemical plants, electronic circuits pose an explosion hazard. **Pneumatic controllers** use compressed air (3–15 psi) as the signal medium.

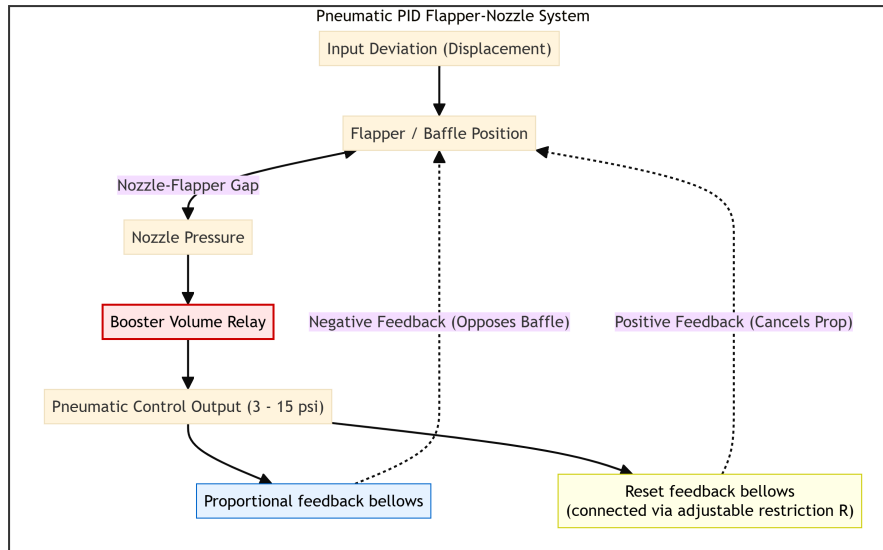


Figure 2.4: Pneumatic PID Controller Architecture

A. Operating Mechanism

- **Flapper-Nozzle System:** The core amplifier. A baffle (flapper) sits close to an open nozzle. If the flapper moves closer to the nozzle, it restricts air escape, increasing the nozzle back-pressure. A flapper movement of just a few micrometers shifts the nozzle pressure across its full range.
- **Booster Relay:** Amplifies the small volume of nozzle back-pressure to provide sufficient air volume to drive the control valve.
- **Feedback Bellows:** To implement PID action, feedback is introduced:
- *Proportional action:* A proportional bellows opposes the flapper movement (negative feedback), reducing sensitivity and establishing a proportional band.
- *Integral (Reset) action:* A reset bellows opposes the proportional bellows (positive feedback). It is connected to the output line through a needle valve (adjustable restriction R). When a deviation occurs, the reset bellows slowly matches the pressure of the proportional bellows, continuously shifting the flapper until the error is zero.

2.3.3 3. Digital PID Controller Implementation

Modern industrial control systems (PLCs, DCS) run digital control algorithms on microprocessors. The continuous PID equation must be discretized.

Let T_s be the sampling interval. The error is sampled as e_k at time step k .

- **Position Algorithm:**

Calculates the absolute control output value (u_k):

$$u_k = K_c \left[e_k + \frac{T_s}{\tau_I} \sum_{i=1}^k e_i + \frac{\tau_D}{T_s} (e_k - e_{k-1}) \right] + u_0$$

- **Velocity Algorithm:**

Calculates the change in controller output ($\Delta u_k = u_k - u_{k-1}$):

$$\Delta u_k = K_c \left[(e_k - e_{k-1}) + \frac{T_s}{\tau_I} e_k + \frac{\tau_D}{T_s} (e_k - 2e_{k-1} + e_{k-2}) \right]$$

- *Advantage:* Does not require initializing a bias value (u_0) and is immune to reset windup because it does not accumulate a running sum of errors.

2.4 Section 4: Controller Tuning Methods [15M] [Priority: High]

Controller tuning is the process of adjusting the controller parameters (K_c , τ_I , τ_D) to obtain the desired closed-loop response.

2.4.1 1. Performance Criteria

The effectiveness of controller tuning is evaluated using transient and steady-state error criteria:

- **Decay Ratio:** The ratio of the amplitude of an oscillation peak to the amplitude of the preceding peak.
- **Quarter-Amplitude Decay:** A standard tuning criterion where the amplitude of each successive peak is reduced by a factor of four (1/4).
- **Integral Error Criteria:**
 - *Integral Absolute Error (IAE):* $\int |e(t)| dt$ (evaluates total absolute deviation; treats small and large errors equally).
 - *Integral Square Error (ISE):* $\int e(t)^2 dt$ (heavily penalizes large errors; used for system safety protection).
 - *Integral Time-weighted Absolute Error (ITAE):* $\int t|e(t)| dt$ (penalizes errors that persist for a long time; produces fast-settling systems).

2.4.2 2. Ziegler-Nichols (ZN) Tuning Methods

A. ZN Method 1: Process Reaction Curve (Open Loop)

This method is used for self-regulating processes that exhibit an S-shaped step response.

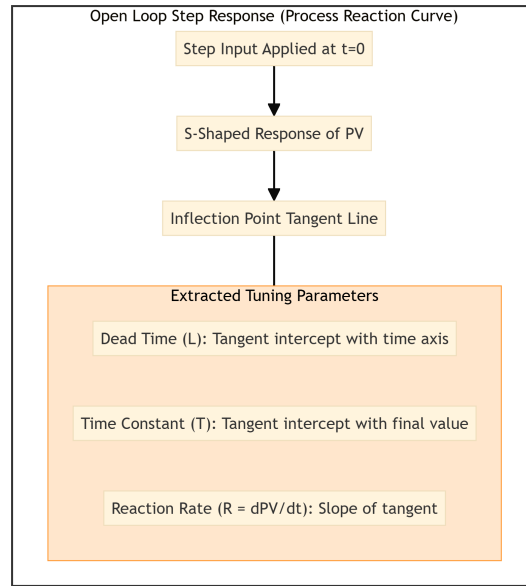


Figure 2.5: Ziegler-Nichols Process Reaction Curve

1. Set the controller to manual mode.
2. Apply a step change of magnitude ΔM to the control valve output.
3. Record the S-shaped response of the process variable.
4. Draw a tangent line at the inflection point of the curve.
5. Extract two parameters:
 - **Dead time (L):** The time intercept where the tangent line meets the initial level.
 - **Time constant (T):** The time required for the process to complete its change along the tangent slope.
 - **Reaction Rate (R):** The slope of the tangent ($R = \Delta PV/T$).

ZN Open-Loop Tuning Table:

Controller Type	K_c	τ_I	τ_D
P	$\frac{T}{L}$	—	—
PI	$0.9 \frac{T}{L}$	$3.3L$	—
PID	$1.2 \frac{T}{L}$	$2.0L$	$0.5L$

B. ZN Method 2: Ultimate Gain Method (Closed Loop)

This method is based on finding the boundary of stability in a closed-loop system.

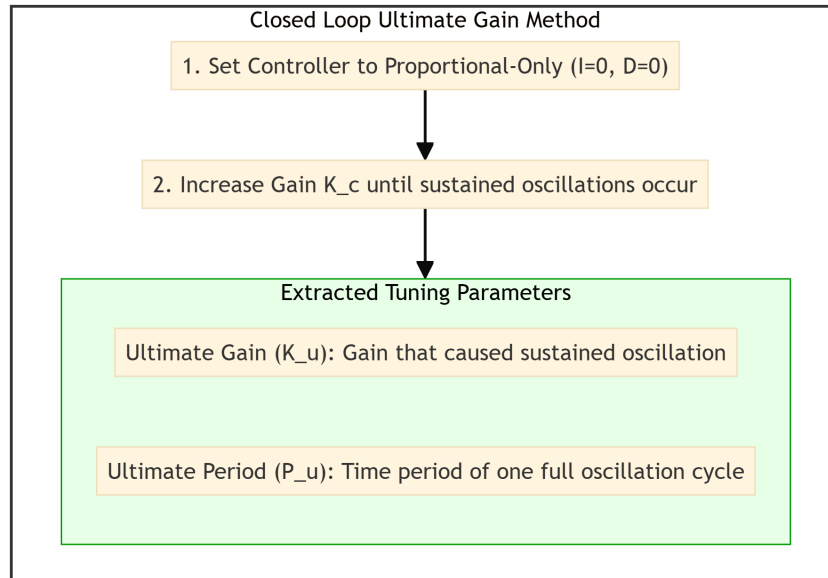


Figure 2.6: Ziegler-Nichols Ultimate Gain Method

1. Connect the controller to the process in closed-loop mode.
2. Turn off the integral and derivative actions ($\tau_I \rightarrow \infty$, $\tau_D = 0$).
3. Set the proportional gain (K_c) to a small value.
4. Introduce a small setpoint disturbance.
5. Gradually increase K_c in steps. At each step, introduce a setpoint disturbance and observe the process variable.
6. Find the **Ultimate Gain** (K_u): the gain value at which the process variable begins to exhibit sustained, constant-amplitude oscillations.
7. Measure the **Ultimate Period** (P_u): the time period of one full oscillation cycle (in minutes or seconds).

ZN Closed-Loop Tuning Table:

Controller Type	K_c	τ_I	τ_D
P	$0.5K_u$	—	—
PI	$0.45K_u$	$\frac{P_u}{1.2}$	—
PID	$0.6K_u$	$0.5P_u$	$\frac{P_u}{8}$

2.4.3 3. Cohen-Coon (CC) Tuning Method

The Cohen-Coon method is an open-loop tuning technique designed for systems with relatively large dead times (L) compared to the time constant (T). It provides better performance than Ziegler-Nichols for processes where the ratio L/T is between 0.6 and 2.0.

- **Parameters:** It uses the same open-loop parameters: process gain (K_p), dead time (L), and time constant (T).
- **Formulas:** The tuning equations are derived to achieve a **quarter-amplitude decay** response:
- *Proportional Gain (K_c):*

$$K_c = \frac{1}{K_p} \frac{T}{L} \left[\frac{4}{3} + \frac{L}{4T} \right] \quad (\text{for P-only controller})$$

- *PID parameters:*

$$K_c = \frac{1}{K_p} \frac{T}{L} \left[\frac{4}{3} + \frac{L}{4T} \right], \quad \tau_I = L \left[\frac{32 + 6(L/T)}{13 + 8(L/T)} \right], \quad \tau_D = L \left[\frac{4}{11 + 2(L/T)} \right]$$

- **Comparison with ZN:** Cohen-Coon accounts for the fractional dead-time (L/T) directly in its gain equations. This makes it less aggressive than Ziegler-Nichols, reducing over-tuning oscillations in dead-time-dominated systems.

2.5 Section 5: Advanced PID Design & Loop Dynamics (Q2.6) [10M][[]]

Implementing PID controllers requires addressing hardware efficiency and understanding the dynamic effects of controller adjustments on process loops.

2.5.1 1. 2-OP-AMP PID Controller Design

A standard parallel PID controller is typically designed using 3 or 4 operational amplifiers (one for each of the P, I, and D stages, and one for summing/inverting). However, using a **2-OP-AMP design** is highly efficient:

- **The Circuit Configuration:**
1. **First OP-AMP Stage (PD with Filter / PI Stage):** A single OP-AMP can combine proportional and derivative action by placing a resistor R_1 in parallel with a capacitor C_1 in the input path, and a parallel resistor-capacitor network (R_f and C_f) in the feedback path. Alternatively, it can form a combined PI stage.
 2. **Second OP-AMP Stage (Summing / Inverting Integrator):** Sums the output of the first stage with the remaining action (e.g., an integrating channel) and acts as an inverting buffer to restore positive gain polarity.
- **Significance:** Reduces component count, power usage, and thermal drift.

2.5.2 2. Steady-State Offset Characteristics

- **Proportional (P) Controller:** Exhibits the **maximum offset**. It requires a permanent error signal ($e_{ss} \neq 0$) to generate a steady-state output different from the default bias p_0 when load shifts occur.
- **Proportional-Integral (PI) and PID Controllers:** Exhibit **zero offset** ($e_{ss} = 0$). The integral term ($K_I \int e \, dt$) accumulates the error over time. Even if the error is tiny, the integral accumulation continues to shift the controller output, resetting the process variable exactly to the setpoint.

2.5.3 3. Overshoot and the Derivative Time Constant (T_d)

- **The Problem:** High overshoot occurs when the process variable approaches the setpoint too rapidly, causing it to cross the target.
- **The Derivative Solution:** The derivative mode acts as a brake. Because it responds to the rate of change of error ($\tau_D \frac{de}{dt}$), it acts in opposition to the fast-rising process variable.
- **Damping Effect:** To reduce overshoot, the **derivative time constant (T_d or τ_D) must be increased**. Increasing T_d increases the damping factor of the closed-loop system, slowing down the rate of approach near the setpoint and settling the system quickly without oscillation.

2.5.4 4. Direct-Acting vs. Reverse-Acting Controllers

- **Direct-Acting Controller:** The controller output **increases** when the process variable (PV) **increases** (or error increases).

Example:* A cooling water loop. If the process temperature (PV) rises, the controller output must increase to open a cooling water valve wider.

- **Reverse-Acting Controller:** The controller output **increases** when the process variable (PV) **decreases** (or error decreases).

Example:* A steam heating loop. If the process temperature (PV) falls, the controller output must increase to open a steam heating valve wider.

2.5.5 5. Pressure Actuator Selection Case Study

Consider a pressure control loop using a proportional-integral (PI) controller with a process time constant of 10 seconds.

- **Actuator Choice:** A **piston and cylinder actuator** is preferred.
- **Why?**
 1. *Speed of Response:* A piston-cylinder design has lower volume and high force-to-mass ratio compared to large pneumatic diaphragms, making its response extremely fast (essential to match the relatively fast 10-second time constant).
 2. *High Pressures:* Pressure loops often operate at high fluid velocities and differential pressures. Piston actuators handle higher pressures and provide longer strokes with stable positioning, preventing valve chatter in PI loops.

Chapter 3

Automation Systems (PLC, DCS, SCADA)

3.1 Section 1: Programmable Logic Controllers (PLC) [15M] [Priority: High]

3.1.1 1. Definition and Architecture

A **Programmable Logic Controller (PLC)** is a solid-state, industrial computer designed to monitor inputs from field devices, execute custom control programs (such as logic, sequencing, timing, and arithmetic), and adjust output devices to automate manufacturing processes in harsh environments.

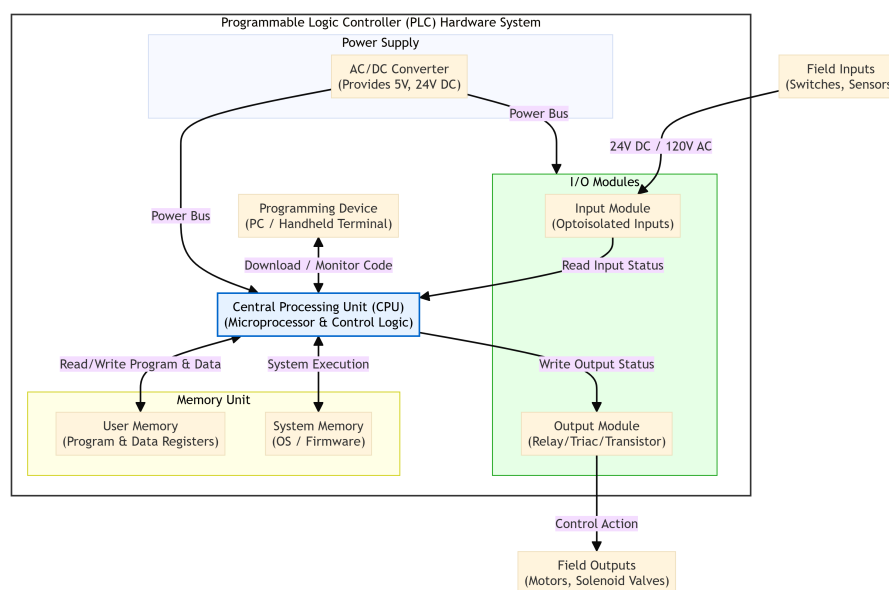


Figure 3.1: PLC Hardware Architecture Block Diagram

Core Architecture Components:

- **Central Processing Unit (CPU):** The brain of the PLC. It reads physical input states, solves the user program sequentially, performs system diagnostics, and updates physical outputs.
- **Memory Unit:**

- **System Memory:** Stores the operating system, communication protocols, and execution firmware.
- **User Memory:** Holds the compiled ladder logic program and data tables (input/output status tables, timer/counter presets, and register values).
- **Power Supply:** Converts industrial AC voltage (120–240 V) or high DC voltage (24 V) into the regulated low-voltage DC (5 V) required to power the internal CPU and memory chips.
- **I/O Modules:** The physical interface to field equipment:
 - **Input Modules:** Convert raw signals from field sensors (e.g., switches, limit sensors) into logic-level signals. They use **optocouplers** to provide galvanic isolation, protecting the CPU from electrical surges.
 - **Output Modules:** Convert low-power CPU decisions into industrial signals (using Relays for AC/DC switching, Triacs for AC loads, or Transistors for high-speed DC switching) to control actuators (e.g., motors, solenoid valves).
- **Programming Device:** A PC running dedicated software used to write, simulate, debug, and download the control logic to the CPU.

3.1.2 2. PLC vs. Conventional Electromagnetic Relay Control Panels

Before PLCs, industrial automation relied on hardwired cabinets filled with electromagnetic relays, timers, and counters.

Feature / Criteria	Hardwired Relay Panels	Programmable Logic Controllers (PLC)
Logic Modification	Requires rewiring, physically running copper cables, and adding relays.	Modified via software on a PC; no physical wiring changes needed.
Footprint & Space	Massive; cabinets occupy significant floor space.	Compact; a single PLC replaces hundreds of physical relays.
Troubleshooting	Extremely difficult; requires checking thousands of wire terminals manually.	Simple; PC screen displays active current flow (highlighted rungs) and faults.
Reliability	Low; mechanical relay contacts wear out, pit, and collect dust.	High; solid-state electronics with no moving parts.
Response Time	Slow (10–50 ms) due to physical armature movement.	Fast (1–10 ms) solid-state execution cycle.
Cost	High wiring labor cost and maintenance downtime.	Low lifetime cost; high initial hardware cost offset by flexibility.

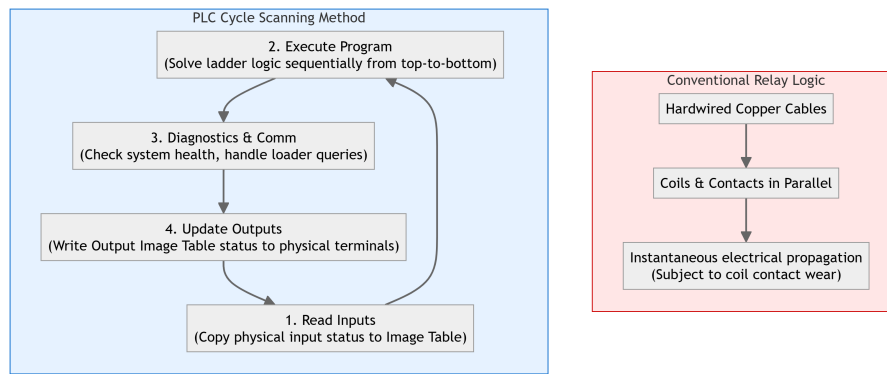


Figure 3.2: Relay Panel vs. PLC Scan Cycle

3.1.3 3. IEC 61131-3 PLC Programming Languages

The international standard IEC 61131-3 defines five standard programming languages for PLCs:

1. **Ladder Diagram (LD):** A graphical language that mimics electrical schematic diagrams. It uses two vertical rails (power and return) and horizontal rungs containing contact switches (NO/NC) and coils, making it highly intuitive for electricians.
2. **Function Block Diagram (FBD):** A graphical language where logic is represented as interconnected blocks containing pre-programmed functions (such as PID, gates, timers). It is popular in process control.
3. **Structured Text (ST):** A high-level, text-based programming language similar to Pascal or C. It is used for complex calculations, array loops, and data manipulation.
4. **Instruction List (IL):** A low-level, assembly-like textual language. It is fast-executing but harder to read and maintain.
5. **Sequential Function Chart (SFC):** A graphical language used to program sequential, multi-step processes. It breaks the logic into steps, transitions, and actions.

3.1.4 4. PLC Ladder Logic Designs

Standard Ladder Diagram (LD) structures use standard contacts: Normally Open (NO: '[]'), Normally Closed (NC: '[]/[]'), and Output Coils ('()').

A. Motor Start/Stop Latching Control

- **Application:** A standard latching (or sealing) circuit allows a momentary push-button to start a motor, which remains running until a stop button is pressed or an overload fault occurs.
- **Circuit Operation:**
 - *Emergency Stop (NC)* and *Stop (NC)* are placed in series at the start of the rung. Their contacts remain closed under normal conditions.
 - *Start (NO)* is placed in series with the *Overload Contact (NC)* and the *Motor Coil (O:0/1)*.

- *Motor Latch Contact (NO)* (addressed to the Motor Coil ‘O:0/1’) is placed in parallel with the *Start (NO)* contact.
- Pressing the *Start* button completes the circuit, energizing the *Motor Coil*.
- When the *Motor Coil* energizes, its auxiliary contact *Motor Latch (NO)* closes. This maintains current flow to the coil even after the *Start* button is released.
- Pressing the *Stop* button or an *Overload* trip opens the series line, de-energizing the coil and opening the latch.

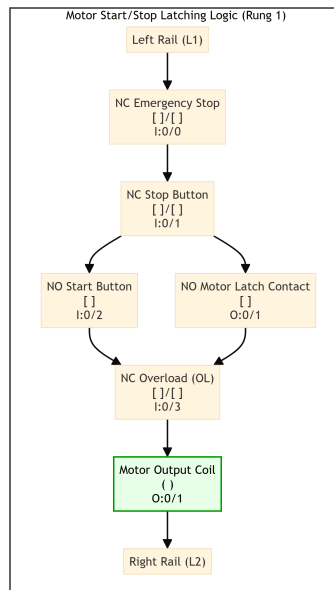


Figure 3.3: Motor Start/Stop Latching Control

B. EX-OR Logic Gate Realization

- **Application:** Realizes the output $Y = A \oplus B = A\bar{B} + \bar{A}B$.
- **Circuit Operation:**
 - *Rung Branch 1:* NO contact A in series with NC contact B .
 - *Rung Branch 2 (in parallel):* NC contact A in series with NO contact B .
 - Both branches lead to the common output coil Y .
 - Output Y is energized if and only if one of the inputs (A or B) is active, but not both.

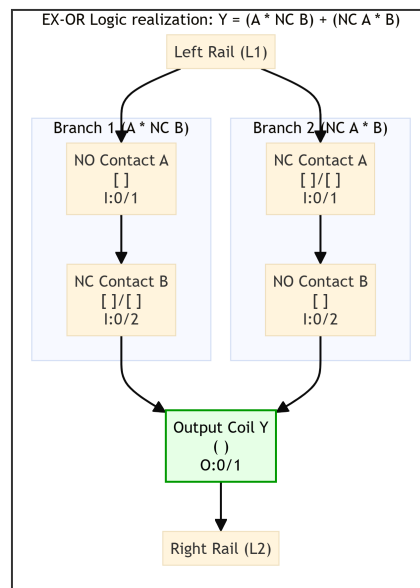


Figure 3.4: PLC EX-OR Gate Implementation

C. Liquid Tank Level Control System

- **Application:** Turn ON the filling pump motor when the liquid level falls to a low-level sensor (L), and keep it running until the level reaches a high-level sensor (H).
- **Circuit Operation:**
 - *Low Sensor L (NO):* Configured to close when the level drops below L .
 - *High Sensor H (NC):* Configured to open when the level reaches H .
 - When the level falls below L , *Low Sensor L* closes, completing the path through the NC *High Sensor H* and energizing the *Pump Coil*.
 - The auxiliary contact *Pump Latch (NO)* closes, sealing the pump output in parallel with *Low Sensor L*.
 - As the tank fills, the level rises above L , opening the *Low Sensor L* contact, but the pump remains energized through the latch.
 - When the level reaches H , the *High Sensor H (NC)* contact opens, breaking the circuit, de-energizing the pump, and resetting the latch.

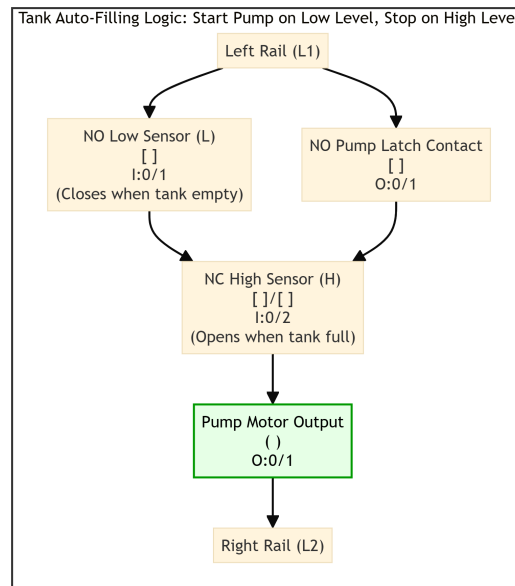


Figure 3.5: PLC Tank Level Control System

D. NAND and NOR Logic Gates

- **NAND Gate** ($Y = \overline{A \cdot B} = \bar{A} + \bar{B}$): Realized by placing NC contact A and NC contact B in parallel. Current flows to the coil if either A is inactive or B is inactive (only cutting power if both are active).
- **NOR Gate** ($Y = \overline{A + B} = \bar{A} \cdot \bar{B}$): Realized by placing NC contact A and NC contact B in series. Current flows to the coil only if both A and B are inactive.

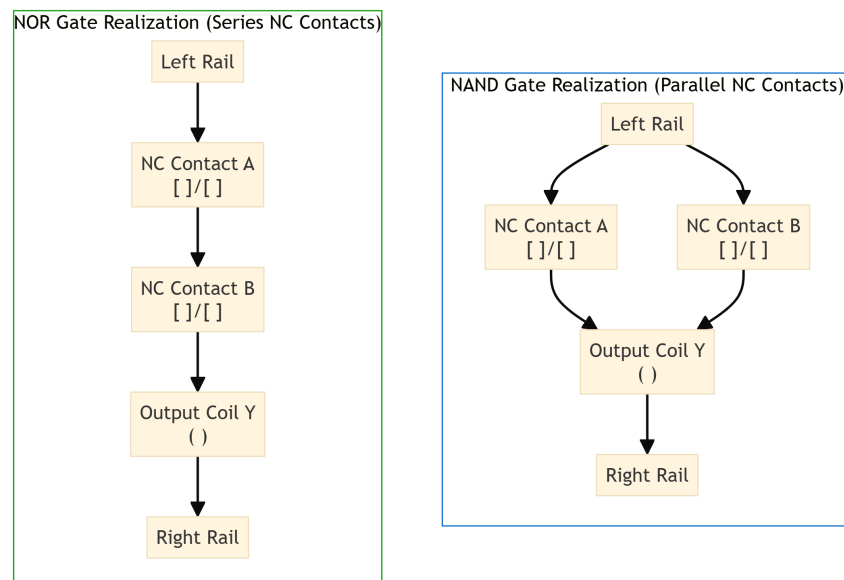


Figure 3.6: PLC NAND & NOR Logic Gate Realizations

3.2 Section 2: Distributed Control Systems (DCS) [15M] [Priority: High]

3.2.1 1. Definition and Importance

A **Distributed Control System (DCS)** is a computerized control system for a large-scale manufacturing plant or continuous process, where autonomous controllers (Local Control Units - LCUs) are distributed throughout the system. The entire system is linked by a high-speed communication network, offering centralized monitoring and engineering management.

Importance in Process Control:

- **No Single Point of Failure:** Because control is distributed among multiple LCUs, a failure in one unit only affects its local plant loop, while the rest of the plant continues operating.
- **Scalability:** Easy to expand by adding new LCUs and Operator Stations to the existing system network.
- **Sophisticated Database & Redundancy:** DCS controllers feature redundant CPUs, power supplies, and network paths to ensure high availability (99.99%).

3.2.2 2. DCS Hierarchical Levels

Industrial automation systems are structured hierarchically into levels as defined by the ISA-95 standard:

- **Level 4: Enterprise Level:** Houses the Enterprise Resource Planning (ERP) database system. It handles corporate-wide logistics, financial decisions, and long-term production scheduling.
- **Level 3: Operations & Management Level:** Manages the Manufacturing Execution System (MES). It oversees plant-wide production scheduling, quality control, maintenance tracking, and database archiving (DCS Historian).
- **Level 2: Supervisory & Console Level:** The interface for operators (HMI consoles) and engineers. It displays real-time process flows, manages alarms, records historical trends, and hosts engineering configuration tools.
- **Level 1: Direct Control Level:** Houses the distributed controllers (LCUs) and local PLCs. These units execute PID control algorithms, check safety interlocks, and calculate analog/digital outputs.
- **Level 0: Field Level:** Comprises physical field instruments (sensors for temperature, pressure, flow) and final control elements (valves, pumps, actuators).

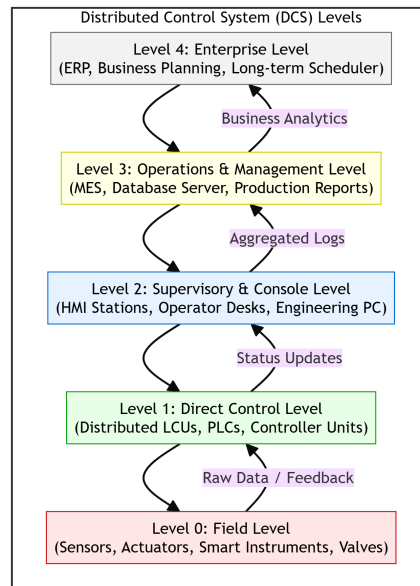


Figure 3.7: DCS Hierarchical Levels

3.2.3 3. DCS Architecture

A standard DCS consists of discrete hardware modules connected via high-speed, redundant network paths.

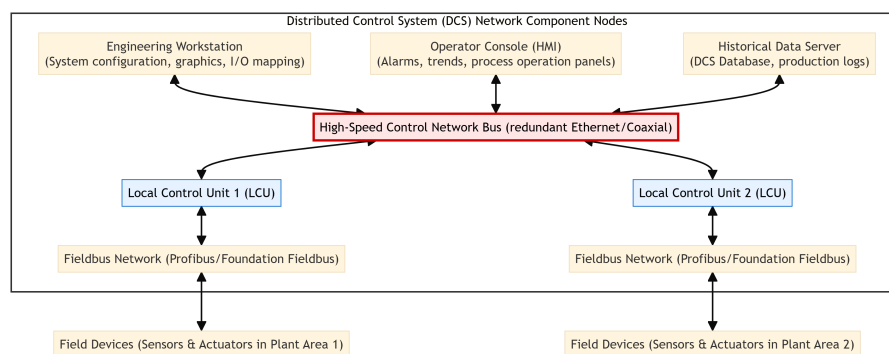


Figure 3.8: DCS Functional Block Architecture

- **Local Control Unit (LCU):** The controller node. It contains a CPU, memory, and local I/O cards to interface with sensors and actuators. It runs the local control algorithms (e.g., PID loops, interlocks).
- **Operator Station (HMI):** Graphic consoles that provide operators with real-time displays of the plant status, alarm lists, trend charts, and manual overrides.
- **Engineering Station:** A dedicated PC used to configure control strategies, design HMI graphics, map I/O registers, and push updates to the LCUs.
- **Historical Data Server:** A high-capacity database node that archives process logs, alarm histories, and system events for analysis.

- **High-Speed System Bus:** A redundant, high-bandwidth network (e.g., Industrial Ethernet, token ring) that carries data between the LCUs, operator stations, and servers.
- **Fieldbus Network:** The local network link (e.g., Profibus, Foundation Fieldbus) connecting smart field instruments directly to the LCU's digital ports.

3.2.4 4. DCS Types and Applications

DCS installations vary based on network architecture and system scope:

- **Traditional Proprietary DCS:** Uses custom, manufacturer-specific hardware, operating systems, and network buses (e.g., Honeywell TDC3000). Highly secure and stable, but expensive and difficult to integrate with third-party components.
- **Open DCS:** Built on standard PC hardware, standard operating systems (Windows/Linux), and open network standards (TCP/IP, Ethernet). Offers lower costs and easy integration, but requires careful cybersecurity management.
- **Application Example:**
 - *Petrochemical Refineries:* Managing hundreds of continuous temperature, pressure, and distillation fraction control loops across wide chemical process units.

3.3 Section 3: Supervisory Control & Data Acquisition (SCADA) [15M] [Priority: High]

3.3.1 1. Definition and Significance

SCADA stands for **Supervisory Control and Data Acquisition**. It is a software-hardware system designed to collect real-time data from remote field locations and provide supervisory control over wide geographic distances (telemetry).

Significance of SCADA:

While a DCS is optimized for high-density, continuous analog control within a single localized facility (e.g., a refinery), SCADA is optimized for monitoring and controlling assets spread across hundreds of kilometers (e.g., oil pipelines, water networks, electricity grids).

3.3.2 2. SCADA System Components

A standard SCADA system contains five main components:

- **Master Terminal Unit (MTU):** The central server located at the control headquarters. It sends command queries to remote field stations, stores the centralized database, runs system diagnostics, and hosts the SCADA software.
- **Remote Terminal Unit (RTU):** A rugged, micro-controller-based device deployed at remote field sites. It gathers local sensor data, converts it to digital telemetry formats, transmits it to the MTU, and executes received MTU control commands.

- **Programmable Logic Controllers (PLCs):** Often deployed alongside or in place of RTUs. PLCs excel at fast, localized discrete logic control, while the RTU handles wide-area communication protocols.
- **Communication WAN:** The telemetry medium linking the MTU to the RTUs (e.g., VHF/UHF radio networks, cellular GPRS/4G links, dial-up telephone networks, satellite links, or fiber optic cables).
- **HMI Console:** Operator display PC screens showing geographical maps, alarms, valve controls, and streaming process variables.

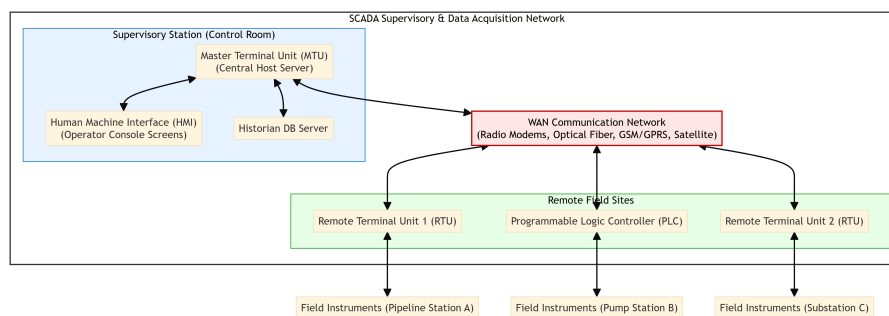


Figure 3.9: SCADA System Topology Block Diagram

3.3.3 3. SCADA Applications

SCADA is widely used in critical infrastructure systems:

- **Electrical Grid Distribution:** Monitoring substation breakers, transformer temperatures, and power line voltage swings across a province.
- **Water & Sewage Networks:** Tracking tank levels, water flow rates, and starting water pumps located at remote wells and reservoir basins.
- **Oil and Gas Pipelines:** Monitoring pressure spikes, flow volumes, and controlling emergency shut-off valves along cross-country pipe pipelines.

3.4 Section 4: Control Valves & Field Accessories [5M] [Priority: Medium]

3.4.1 1. Control Valve Anatomy

A **control valve** is the final control element in a liquid or gas process control loop. It adjusts the fluid flow rate by varying the opening size of the flow path based on a controller output signal.

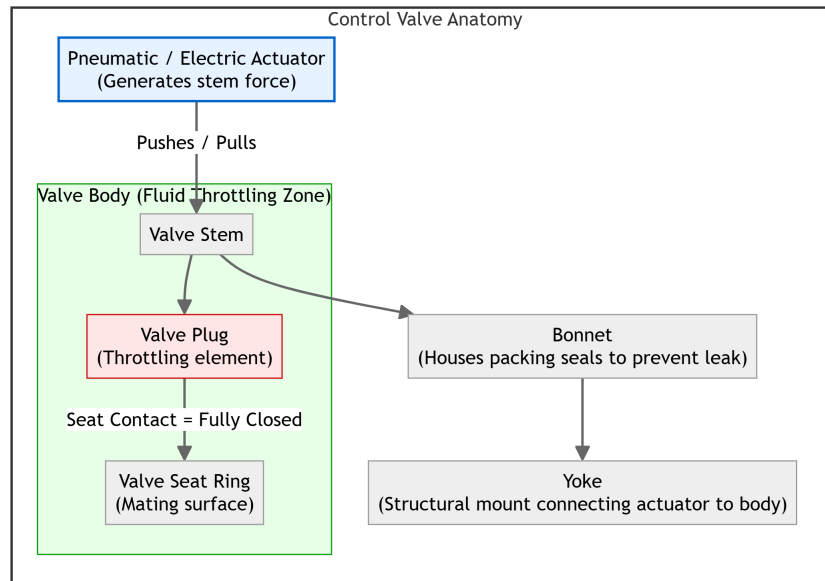


Figure 3.10: Control Valve Structural Component Assembly

Core Components:

- **Actuator:** The drive mechanism that converts the control signal (pneumatic air pressure or electric voltage) into linear or rotary motion to move the valve stem.
- **Yoke:** A metal bracket that securely mounts the actuator housing to the valve body bonnet.
- **Stem:** A polished metal rod that connects the internal valve plug to the external actuator piston. It transmits the actuator displacement.
- **Bonnet:** The top cap of the valve body. It houses the **packing box** (sealing material like Teflon or graphite) to prevent process fluid from leaking along the moving stem.
- **Valve Plug:** The movable throttling element inside the valve body. Its position relative to the seat determines the flow area.
- **Valve Seat Ring:** The stationary ring fixed inside the valve body. When the valve plug is fully pressed against the seat, the flow path is completely blocked.
- **Valve Body:** The outer pressure vessel shell that contains the internal plug and seat, managing the fluid flow path.

3.4.2 2. Close and Open States

- **Fully Closed State:** When the actuator pushes the stem until the valve plug is in full contact with the **valve seat ring**. This blocks all fluid flow.
- **Fully Open State:** When the actuator retracts the stem, lifting the valve plug to its maximum distance from the valve seat ring, allowing maximum flow.
- **Throttling State:** Operating at intermediate positions between fully open and fully closed to maintain a specific process flow rate.

3.4.3 3. Globe Valves

A **globe valve** is a linear-motion control valve commonly used for throttling flow control.

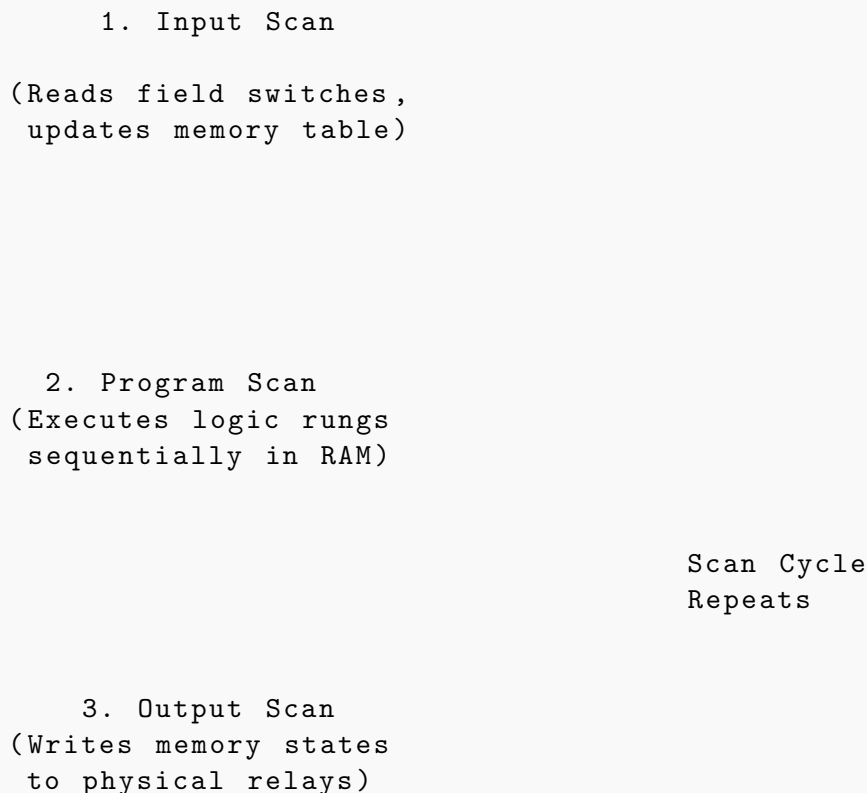
- **Operation:** It has a spherical body shape with an internal baffle partition. The flow path is forced to turn 90° twice as it passes through the orifice. A plug on the end of a linear stem is lowered or raised to restrict or permit flow.
- **Application:** Globe valves are ideal for throttling pressure and flow control because they offer precise flow control and can withstand high differential pressures, though they introduce a significant pressure drop across the valve.

3.5 Section 5: PLC Performance, Noise Immunity, and Instruction Sets (Q3.5) [5M][[]

High-reliability industrial operation requires an understanding of how PLCs manage CPU timing, survive harsh electric noise environments, and execute basic timer logic.

3.5.1 1. PLC Scan Time

The **Scan Time** of a PLC is the total amount of time required for the CPU to execute one complete cycle of the operating program. This cycle is performed continuously and sequentially.



```
4. Housekeeping/Comm  
(Self-diagnostics &  
  
HMI communication)
```

Steps of the PLC Scan Cycle:

1. **Input Scan:** The CPU reads the ON/OFF states of all physical input modules (e.g., push buttons, sensors) and copies these values into a dedicated memory sector called the **Input Image Table**.
2. **Program execution (Logic Scan):** The CPU reads the ladder logic program sequentially from top to bottom (rung by rung, left to right). It resolves all contacts using the states stored in the Input Image Table and writes the updated output states to the **Output Image Table**.
3. **Output Scan:** The CPU copies the states from the Output Image Table directly to the physical output hardware modules to energize or de-energize external actuators (e.g., solenoids, contactors).
4. **Housekeeping & Communications:** The CPU checks its internal hardware for diagnostics, updates its internal timers, and handles communications with programming PCs or HMI consoles.

3.5.2 2. PLC Noise Immunity

Industrial plants are filled with **Radio Frequency Interference (RFI)**, electromagnetic fields (EMI), and voltage spikes from high-power switches. PLCs offer **excellent noise immunity** compared to standard microcontrollers or old relay setups through several hardware-based design choices:

1. **Optical Isolation (Galvanic Isolation):** Physical inputs and outputs are isolated from the CPU using **optocouplers**. The incoming electrical signal drives an internal LED. The light travels across an air gap to a phototransistor, converting the signal back to low-voltage CPU logic. There is no physical electrical connection, preventing high-voltage spikes (surges up to 5 kV) from frying the processor.
2. **Input Filtering Circuits:** Every input channel features an RC low-pass filter to block high-frequency electromagnetic noise spikes and filter out switch contact bounce.
3. **Faraday Caging & Shielding:** PLCs are enclosed in heavy-duty grounded metal or high-impact plastic enclosures to block electrostatic fields. Internal buses are shielded to prevent cross-talk.
4. **Reliability Comparison:**

- **Conventional Relay Panels:** High mechanical noise, contact arcing, and pitting.
- **Microcontrollers / PCs:** Poor noise immunity; high susceptibility to electrostatic discharges (ESD) and line ripples which trigger system resets or data corruption.
- **PLCs:** Excellent immunity, designed to withstand continuous industrial environment hazards.

3.5.3 3. Timer On Delay (TON) Instruction

The **Timer On Delay (TON)** is a standard PLC instruction used to delay turning on an output. When the input rung preceding the TON block transitions from False to True, the timer begins accumulating time.

Timer On Delay (TON)

Timer:	T4:0
Time Base:	1.0s
Preset (PRE):	10
Accumulator (ACC):	0

Core Parameters:

- **Timer Address:** The memory register identifier (e.g., 'T4:0').
- **Time Base:** The resolution of the timer, typically 0.01 s, 0.1 s, or 1.0 s.
- **Preset Value (PRE):** The target delay time.
- **Accumulator Value (ACC):** The elapsed time counted so far.
- **Control Bits:**
 1. **Enable Bit (EN):** Set to 1 as long as the input rung is active (True).
 2. **Timer Timing Bit (TT):** Set to 1 while the timer is actively counting ($ACC < PRE$ and $EN = 1$).
 3. **Done Bit (DN):** Set to 1 when $ACC \geq PRE$. This bit is used as a contact to trigger the delayed physical output.

State Transitions:

- **Rung goes True:** EN becomes 1, TT becomes 1, and ACC increments every time base interval.
- **ACC reaches PRE:** TT drops to 0 and DN becomes 1. The connected output device turns ON.
- **Rung goes False:** EN, TT, and DN immediately drop to 0. The ACC resets back to 0 (this is a non-retentive behavior).

3.5.4 4. Major Industrial PLC Manufacturers

To design and install systems, engineers choose PLCs from established hardware vendors:

- **Siemens (Germany):** Standard-setter in Europe; famous for the Simatic S7 series (S7-1200, S7-1500) and TIA Portal software.
- **Allen-Bradley / Rockwell Automation (USA):** Market leader in North America; famous for ControlLogix, CompactLogix, and RSLogix/Studio 5000 software.
- **Mitsubishi Electric (Japan):** Dominant in Asian manufacturing; famous for the MELSEC series (FX, Q-series).
- **ABB (Switzerland):** Widely used in process automation and power grid systems; famous for the AC500 and AC800M controllers.
- **Schneider Electric (France):** Renowned for its Modicon line (the original creators of the PLC).

*(Note: **Microsoft** is a software/operating system developer and does **not** manufacture industrial PLCs).*

Chapter 4

Advanced Control Techniques

4.1 Section 1: Feedforward & Ratio Control [15M] [Priority: High]

4.1.1 1. Feedforward Control System

A **feedforward control system** is an anticipatory control strategy where disturbances are measured and corrected for before they have a chance to affect the controlled process variable.

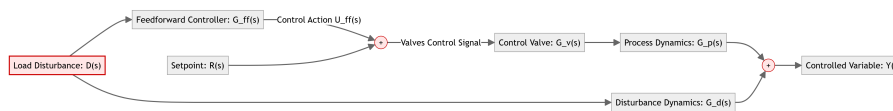


Figure 4.1: Feedforward Control System Block Diagram

A. Mathematical Derivation of Feedforward Controller

Consider the block diagram of the process system. The controlled output $Y(s)$ is affected by both the control action $U(s)$ and the load disturbance $D(s)$:

$$Y(s) = G_p(s)G_v(s)U(s) + G_d(s)D(s)$$

where:

- $G_p(s)$: Process dynamics transfer function.
- $G_v(s)$: Control valve transfer function.
- $G_d(s)$: Disturbance dynamics transfer function.

The feedforward controller $G_{ff}(s)$ measures the disturbance $D(s)$ and generates the control signal $U_{ff}(s) = G_{ff}(s)D(s)$. For perfect control, we want to maintain the output at zero deviation ($Y(s) = 0$) despite changes in $D(s)$:

$$G_p(s)G_v(s)[G_{ff}(s)D(s)] + G_d(s)D(s) = 0$$

Dividing by $D(s)$:

$$G_p(s)G_v(s)G_{ff}(s) + G_d(s) = 0$$

Solving for the ideal Feedforward Controller Transfer Function $G_{ff}(s)$:

$$G_{ff}(s) = -\frac{G_d(s)}{G_v(s)G_p(s)}$$

(Note: The negative sign indicates that the controller must apply a corrective action in the opposite direction of the disturbance deflection).

B. Feedback vs. Feedforward Control Comparison

Feature / Criteria	Feedback Control	Feedforward Control
Operating Principle	Reactive; acts only after a deviation in the output occurs.	Proactive; acts before a deviation in the output occurs.
Measurement Parameter	Measures the controlled variable ($Y(s)$).	Measures the load disturbance ($D(s)$).
Stability	Can cause instability and oscillations if tuned aggressively.	Inherently stable because it does not introduce a feedback loop.
Process Knowledge	Requires minimal process model knowledge.	Requires an accurate mathematical model of the process.
Disturbance Rejection	Rejects all disturbances, both known and unknown.	Rejects only the specific, measured disturbances.

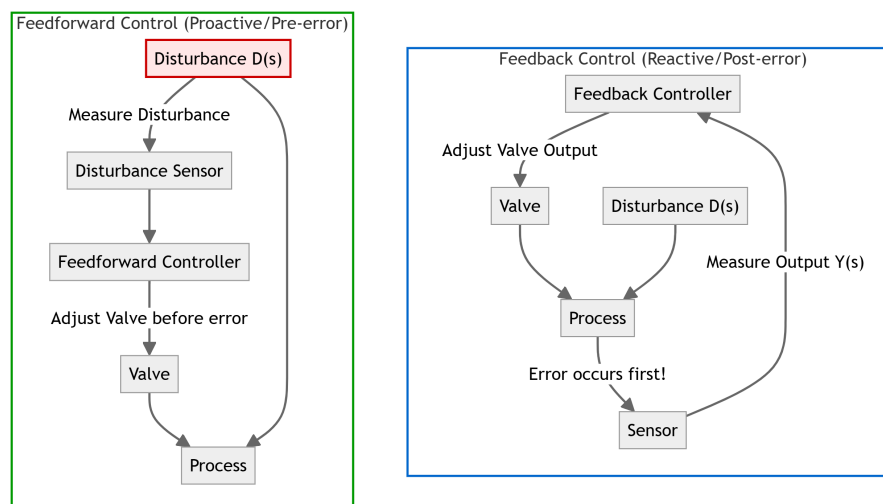


Figure 4.2: Feedback vs. Feedforward Control Loop Comparison

C. Heat Exchanger Case Study

In an industrial shell-and-tube heat exchanger, process fluid is heated by steam.

- **Goal:** Maintain process fluid outlet temperature at setpoint.
- **Load Disturbances:** Inlet process fluid flow rate, inlet process fluid temperature, and steam pressure variations.

- **FF Control Action:** A sensor measures the inlet flow rate (disturbance). If the flow rate increases, the feedforward controller immediately opens the steam control valve to supply more heat before the outlet temperature has a chance to drop.

4.1.2 2. Ratio Control System

A **ratio control system** is a specialized feedforward control loop used to maintain a constant ratio between two process flow streams.

A. Wild Flow vs. Controlled Flow

- **Wild Flow (F_W):** The primary stream flow rate. It is determined by upstream plant conditions and cannot be adjusted by the ratio controller.
- **Controlled Flow (F_C):** The secondary stream flow rate. Its control valve is adjusted by the controller to maintain the desired ratio:

$$R = \frac{F_C}{F_W}$$

B. Multiplier vs. Divider Schemes

- **Multiplier Scheme (Recommended):**

The wild flow F_W is measured and multiplied by the desired ratio R to calculate the setpoint for the controlled flow loop ($F_C^* = R \times F_W$).

- **Divider Scheme:**

Both flows are measured, and their ratio is calculated ($R_{act} = F_C/F_W$) and compared to the setpoint ratio R .

- **Disadvantage:** If the wild flow drops to zero ($F_W = 0$), the division results in a divide-by-zero error, causing controller saturation and loop instability. Therefore, the **multiplier scheme** is preferred.

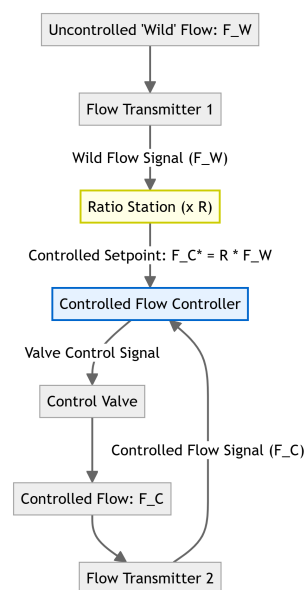


Figure 4.3: Ratio Control System (Multiplier Scheme)

- **Application Example:**

- *Fuel-Air Ratio Control:* Maintaining the exact stoichiometric ratio of air and fuel entering a combustion furnace to ensure efficient combustion and reduce emissions.

4.2 Section 2: Cascade Control System [15M] [Priority: High]

4.2.1 1. Definition and Operating Principle

A **cascade control system** consists of two nested feedback loops: a **primary (outer) loop** and a **secondary (inner) loop**. The primary controller (Master) monitors the main controlled variable and calculates a control output, which serves as the setpoint for the secondary controller (Slave). The slave controller directly adjusts the final control valve.

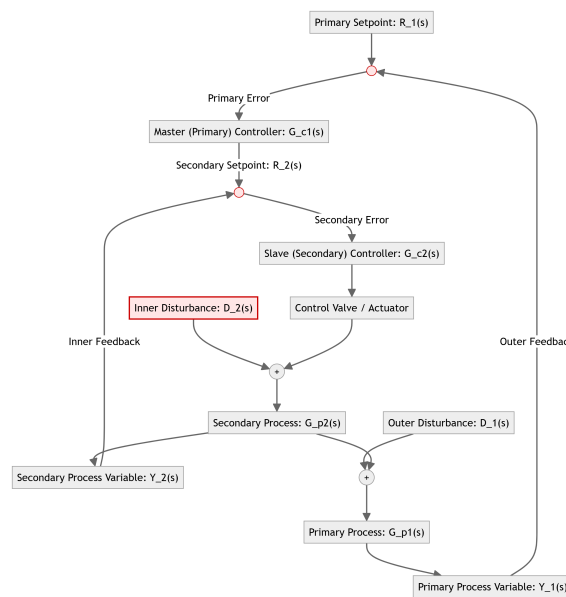


Figure 4.4: Cascade Control Loop Block Diagram

Operating Principle:

- **Inner (Slave) Loop:** Deployed close to the source of the disturbance (e.g., fuel header pressure). It is designed to be fast-acting.
- **Outer (Master) Loop:** Monitors the primary slow process variable (e.g., furnace temperature). It calculates the setpoint for the inner loop.
- If a pressure spike occurs in the fuel line, the slave loop detects the change and corrects the valve position immediately, before the primary process temperature is affected.

4.2.2 2. Advantages of Cascade Control

- **Fast Disturbance Rejection:** Rejects secondary disturbances within the inner loop before they can propagate and affect the primary process variable.

- **Reduced Phase Lag:** Breaking a multi-capacity process into two smaller loops reduces the overall phase lag, allowing higher controller gains and faster response.
- **Linearization:** The inner loop linearizes the valve characteristics and compensates for actuator non-linearities like hysteresis and stiction.

4.3 Section 3: Special Loop Architectures (Split-Range & Override) [15M] [Priority: High]

4.3.1 1. Split-Range (Duplex) Control

Split-range control is a loop configuration where a single controller output (CO) is split to control two or more final control elements (valves) acting in sequence.

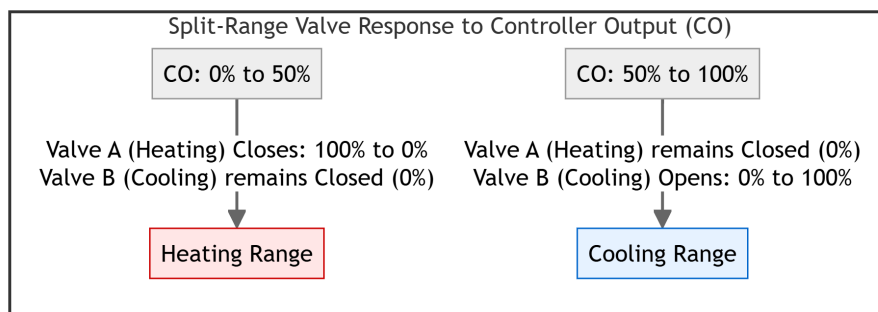


Figure 4.5: Split-Range (Duplex) Control Loop Graph

A. Working Principle

The controller output range (0–100%) is split into sub-ranges for each valve:

- **Valve A (Heating Valve - Air-to-Open):**
 - At 0% CO: Fully Open (3 psi signal).
 - At 50% CO: Fully Closed (9 psi signal).
 - At 50–100% CO: Remains Closed.
- **Valve B (Cooling Valve - Air-to-Close):**
 - At 0–50% CO: Remains Closed (9 psi signal).
 - At 50% CO: Fully Closed.
 - At 100% CO: Fully Open (15 psi signal).

B. Industrial Example

• Chemical Jacketed Reactor:

To control reactor temperature, a single PID controller adjusts both steam (heating) and cooling water (cooling) valves. From 0–50% CO, the steam valve closes. From 50–100% CO, the cooling valve opens. This ensures both valves are never open at the same time, preventing energy waste.

4.3.2 2. Override (Selective) Control

Override (selective) control is a protective control strategy where two or more controllers share a single control valve through a selector block (High Selector or Low Selector).

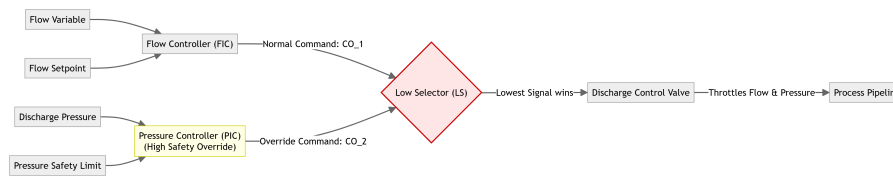


Figure 4.6: Override (Selective) Control Loop

A. Working Principle & Safety Protection

- Under normal operating conditions, the primary controller (e.g., Flow Controller) maintains the process variable at the setpoint.
- If a secondary variable (e.g., discharge pressure or motor current) exceeds a safety threshold, the safety override controller activates.
- The **Low Selector (LS)** or **High Selector (HS)** compares the controller outputs and routes the safer signal to the valve:
- *Low Selector*: Passes the lowest controller output. Used to limit high pressures or flows.
- *High Selector*: Passes the highest controller output. Used to maintain minimum flow to prevent pump damage.

B. Necessity

- Prevents equipment damage (e.g., pipeline overpressure or pump cavitation).
- Avoids plant shutdowns by temporarily throttling the process instead of tripping the system.

4.4 Section 4: Advanced Adaptive & Model-Based Control [15M] [Priority: High]

4.4.1 1. Adaptive Control

An **adaptive controller** automatically adjusts its tuning gains (K_c , τ_I , τ_D) in real-time to compensate for changes in process dynamics, non-linearities, or environmental conditions.

Feature / Criteria	Model Reference Adaptive Control (MRAC)	Self-Tuning Regulators (STR)
Operating Loop	Uses an explicit reference model in parallel.	Identifies process parameters online using recursive estimation.
Parameter Update	Updates controller parameters based on model tracking error.	Updates controller parameters by solving design equations.
Complexity	Simpler calculation loop; relies on direct gradient rules.	Highly complex; requires online parameter estimation.
Response Speed	Fast adaptation to transient path deviations.	Slower adaptation due to parameter identification lag.

A. Model Reference Adaptive Control (MRAC)

An ideal reference model defines the desired response of the closed-loop system to a setpoint change. The adjustment mechanism compares the actual process output (Y) with the model output (Y_m) and updates the controller gains using adaptation algorithms (e.g., MIT rule) to drive the error to zero.

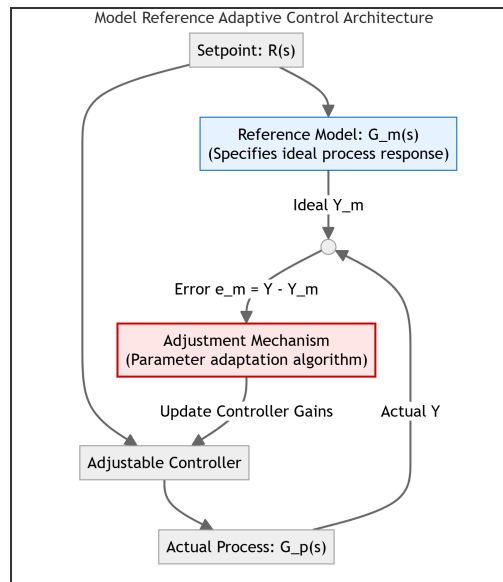


Figure 4.7: Model Reference Adaptive Control (MRAC)

B. Self-Tuning Regulators (STR)

An STR contains an online parameter estimator that continuously identifies the process parameters (e.g., using Recursive Least Squares - RLS). A controller design block uses these estimated parameters to calculate new controller gains (e.g., using pole placement) and updates the controller.

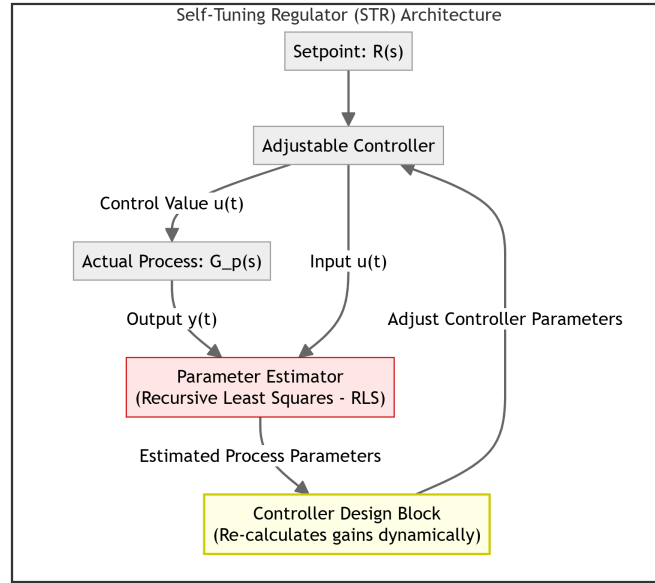


Figure 4.8: Self-Tuning Regulator (STR)

4.4.2 2. Internal Model Control (IMC)

Internal Model Control (IMC) is a model-based control strategy where an explicit mathematical model of the process is run in parallel with the physical process.

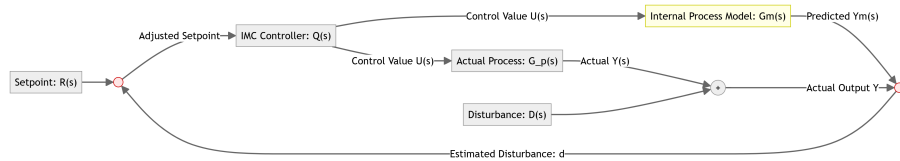


Figure 4.9: Internal Model Control (IMC) Block Diagram

A. Working Principle

- The controller output is sent to both the physical process $G_p(s)$ and the process model $\tilde{G}_p(s)$.
- The predicted model output ($\tilde{Y}$) is subtracted from the actual measured process output (Y). This difference represents the unmodeled disturbances and model mismatch ($d = Y - \tilde{Y}$).
- This difference is fed back to the setpoint summing junction.
- **Advantage:** If the model matches the process exactly ($\tilde{G}_p = G_p$) and there are no disturbances, the feedback signal is zero, and the control loop operates in an open-loop fashion, improving system stability. The IMC structure also simplifies the design of controllers for processes with dead time.

Chapter 5

Quick Revision: PE-EC802B - Industrial Automation and Control

This document compiles high-density definitions, essential formulas, tuning tables, practicing diagrams, and a comprehensive database of solved exam one-liners across all 4 units for active recall.

5.1 Section 1: Core Definitions to Memorize

5.1.1 Unit I: Sensors & Actuators

- **Active Transducers:** Self-generating sensors that convert physical energy directly into electrical energy without requiring an external power source (e.g., Thermocouples, Piezo-electric crystals).
- **Passive Transducers:** Sensors that cause a change in electrical parameters (R, L, C) and require an auxiliary external electrical source to operate (e.g., RTDs, LVDTs, Strain Gauges).
- **LVDT (Linear Variable Differential Transformer):** A passive inductive transducer consisting of a primary coil and two series-opposing secondary coils. It measures linear displacement with infinite resolution by tracking the position of a moving ferromagnetic core.
- **Seebeck Effect:** The generation of an electromotive force (EMF) when two dissimilar metal junctions are held at different temperatures.
- **Gauge Factor (G):** A measure of the sensitivity of a strain gauge, representing the ratio of fractional change in resistance to longitudinal strain.
- **Back EMF (E_b):** The internal voltage induced in a DC motor's rotating armature that opposes the applied terminal voltage.
- **Motor Starters:** Devices that limit the high starting current of a motor (which occurs because Back EMF is zero at standstill) by inserting a variable series resistance.
- **Pneumatic Actuator:** An actuator that converts compressed air signals (standard 3–15 psi) into linear or rotary displacement.

- **Optoisolation:** The use of an optocoupler (LED + phototransistor) to transmit signals across an optical path, preventing high voltage transients on the field side from damaging the controller.
- **Calibration:** The process of comparing an instrument's output against a traceable reference standard of higher accuracy to detect deviations and establish correction bounds.
- **Systematic Errors:** Predictable, unidirectional measurement offsets caused by calibration errors, environmental changes, or parallax. They can be eliminated by calibration.
- **Random Errors:** Unpredictable variations caused by electrical noise or environmental fluctuations. They cannot be eliminated but are reduced statistically using the mean of multiple readings.
- **I/P Converter (Current-to-Pneumatic):** An electro-pneumatic transducer that converts a standard 4–20 mA electrical command current into a proportional 3–15 psi compressed air signal to drive pneumatic actuators.
- **P/I Converter (Pneumatic-to-Current):** A transducer that converts a standard 3–15 psi pneumatic pressure signal into a proportional 4–20 mA DC electrical current signal for controller input.

5.1.2 Unit II: Controller Tuning

- **Servo Operation:** A control loop operating condition where the primary goal is to track a dynamically changing setpoint ($R(s)$).
- **Regulatory Operation:** A control loop operating condition where the primary goal is to keep the process variable constant by rejecting load disturbances ($D(s)$).
- **Self-Regulating System:** A process that naturally finds a new stable equilibrium following a step change in input without any controller intervention (e.g., gravity-drained liquid tank).
- **Non-Self-Regulating (Integrating) System:** A process that continuously drifts or ramps following a step input change until a physical limit is hit (e.g., pump-drained tank).
- **Dead-Time (Transport Lag):** The time delay between the application of an input signal and the first observable process response, modeled as e^{-sT_d} .
- **Proportional Band (PB):** The percentage of error required to drive the controller output from 0% to 100%. Formulated as $PB = 100/K_c$.
- **Reset Windup:** The saturation of an Integral controller's output due to a persistent error, leading to large overshoot as the controller takes time to decrease its integrated value.
- **Noise Amplification:** A major limitation of Derivative controllers where high-frequency measurement noise is amplified by the derivative term ($\tau_D \frac{de}{dt}$), causing control valve wear.
- **Direct-Acting Controller:** A controller configuration where the output increases as the process variable (PV) increases (e.g., cooling loop).
- **Reverse-Acting Controller:** A controller configuration where the output increases as the process variable (PV) decreases (e.g., heating loop).

5.1.3 Unit III: Automation Systems

- **PLC (Programmable Logic Controller):** A rugged, solid-state industrial computer designed to monitor inputs, execute sequential program logic, and control field actuators.
- **PLC Scan Cycle:** The continuous execution cycle consisting of: (1) Read inputs to image table, (2) Solve ladder program sequentially, (3) Perform diagnostics/comm, and (4) Update output terminals.
- **DCS (Distributed Control System):** A control architecture where autonomous controllers (LCUs) are distributed throughout a plant, coordinated via a high-speed system bus, eliminating a single point of failure.
- **SCADA (Supervisory Control and Data Acquisition):** A software-hardware system designed for supervisory control and real-time telemetry data collection from geographically dispersed remote assets.
- **Master Terminal Unit (MTU):** The central host server in a SCADA system that queries remote field stations, stores databases, and drives operator HMIs.
- **Remote Terminal Unit (RTU):** A rugged remote microcontroller deployed at SCADA field sites to collect sensor data, format telemetry, and execute supervisory commands.
- **Globe Valve:** A linear control valve with a spherical body and baffle plate, widely used for throttling fluid pressure and flow rate control.
- **Valve Seat Ring:** The stationary seat ring fixed inside a control valve. When the plug is fully connected to the seat, the valve is fully closed.
- **PLC Scan Time:** The total time required for the CPU to perform one complete cycle of reading physical inputs, executing the user logic program, performing diagnostics, and writing physical outputs.
- **Timer On Delay (TON):** A standard PLC instruction that delays turning on an output (energizing its Done 'DN' bit) for a preset duration after the timer's rung conditions become True.

5.1.4 Unit IV: Advanced Control Techniques

- **Feedforward Control:** A proactive control strategy where load disturbances are measured and corrected for before they affect the process variable.
- **Ratio Control:** A control system designed to maintain a constant ratio between two flow streams ($R = F_C/F_W$).
- **Cascade Control:** A nested loop configuration containing a primary (outer) Master loop and a fast-acting secondary (inner) Slave loop.
- **Split-Range Control:** A loop configuration where a single controller output is split to drive two or more control valves sequentially (e.g. heating and cooling).
- **Override (Selective) Control:** A safety loop where multiple controllers share a single control valve through High/Low Selectors to protect equipment during limit crossings.

- **Adaptive Control:** A control system that adjusts its gains dynamically in real-time to compensate for process gain drift.
- **Internal Model Control (IMC):** A control structure running a process model in parallel with the actual plant to isolate disturbances ($d = Y - \tilde{Y}$) and feed them back to the controller.

5.2 Section 2: Key Formulas & Parameter Tables

5.2.1 1. Transducer & Actuator Physics

- **Thermocouple EMF:**

$$E = a(T_{hot} - T_{cold}) + b(T_{hot} - T_{cold})^2$$

- **Cold Junction Compensation:**

$$V_{net} = V_{raw} + V_{comp}$$

- **RTD Resistance:**

$$R_T = R_0[1 + \alpha(T - T_0)]$$

- **Strain Gauge Gauge Factor (G):**

$$G = \frac{\Delta R/R}{\Delta L/L} = 1 + 2\nu + \frac{\Delta \rho/\rho}{\Delta L/L}$$

where ν is Poisson's ratio, and $\frac{\Delta \rho}{\rho}$ is the piezoresistive effect.

- **Ultrasonic Distance (ToF):**

$$D = \frac{v \times t}{2}$$

where v is the speed of sound (343 m/s at 20°C), and t is the time-of-flight.

- **DC Motor Armature Current & Back EMF:**

$$E_b = V - I_a R_a \implies I_{start} = \frac{V}{R_a} \quad (\text{since } E_b = 0 \text{ at } N = 0)$$

5.2.2 2. Controller Equations

- **Active Low-Pass Filter (LPF) Transfer Function & Cut-off:**

$$H(s) = \frac{A_v}{1 + sRC}, \quad f_c = \frac{1}{2\pi RC}$$

- **Active High-Pass Filter (HPF) Transfer Function & Cut-off:**

$$H(s) = \frac{A_v sRC}{1 + sRC}, \quad f_c = \frac{1}{2\pi RC}$$

- **Parallel PID Controller Transfer Function:**

$$G_c(s) = K_c \left[1 + \frac{1}{\tau_I s} + \tau_D s \right]$$

- **Ideal Feedforward Controller:**

$$G_{ff}(s) = -\frac{G_d(s)}{G_v(s)G_p(s)}$$

- **First-Order Padé Approximation of Dead Time:**

$$e^{-sT_d} \approx \frac{1 - \frac{T_d}{2}s}{1 + \frac{T_d}{2}s}$$

5.2.3 3. Controller Tuning Tables

ZN Open-Loop (Process Reaction Curve) Tuning Table:

- Parameters: Dead time (L), Time constant (T).

Controller Type	Proportional Gain (K_c)	Integral Time (τ_I)	Derivative Time (τ_D)
P	$\frac{T}{L}$	—	—
PI	$0.9\frac{T}{L}$	$3.3L$	—
PID	$1.2\frac{T}{L}$	$2.0L$	$0.5L$

ZN Closed-Loop (Ultimate Gain) Tuning Table:

- Parameters: Ultimate Gain (K_u), Ultimate Period (P_u).

Controller Type	Proportional Gain (K_c)	Integral Time (τ_I)	Derivative Time (τ_D)
P	$0.5K_u$	—	—
PI	$0.45K_u$	$\frac{P_u}{1.2}$	—
PID	$0.6K_u$	$0.5P_u$	$\frac{P_u}{8}$

Cohen-Coon PID Formulas (Quarter-Amplitude Decay):

- Parameters: Process Gain (K_p), Dead time (L), Time constant (T).

$$K_c = \frac{1}{K_p} \frac{T}{L} \left[\frac{4}{3} + \frac{L}{4T} \right], \quad \tau_I = L \left[\frac{32 + 6(L/T)}{13 + 8(L/T)} \right], \quad \tau_D = L \left[\frac{4}{11 + 2(L/T)} \right]$$

5.2.4 4. Error Propagation

- **Addition or Subtraction ($X = x_1 \pm x_2$):**

$$\Delta X = \Delta x_1 + \Delta x_2 \quad (\text{absolute errors sum})$$

- **Multiplication or Division ($X = x_1 \cdot x_2$ or $X = x_1/x_2$):**

$$\frac{\Delta X}{X} = \frac{\Delta x_1}{x_1} + \frac{\Delta x_2}{x_2} \quad (\text{relative errors sum})$$

5.3 Section 3: Diagrams to Practice Sketching

Prepare to sketch these block architectures, response curves, and circuit schematics by hand:

5.3.1 1. LVDT Electro-Mechanical Schematic (Unit I)

Detailed diagram showing the primary and secondary series-opposing windings with a movable magnetic core.

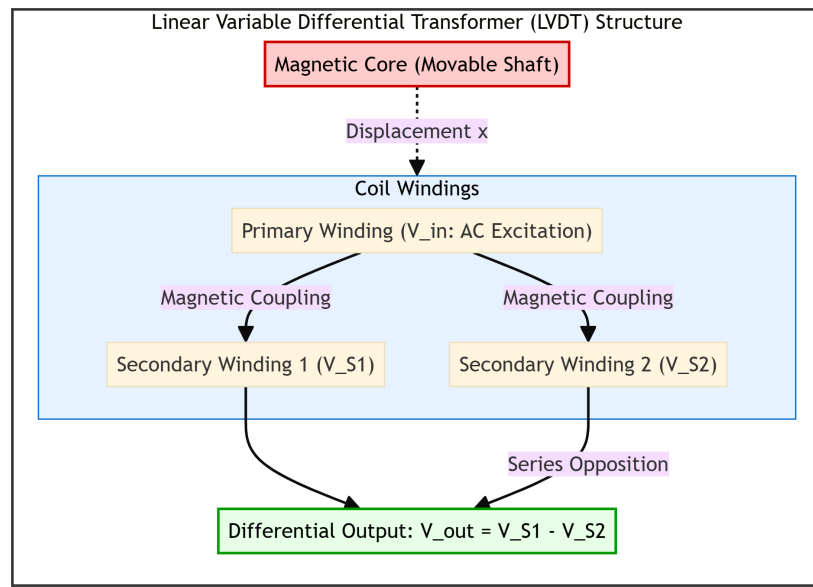


Figure 5.1: LVDT Electro-Mechanical Schematic

5.3.2 2. LVDT Transfer Characteristics (Unit I)

Output voltage magnitude showing a linear response with a null residual voltage, and the 180° phase shift through null.

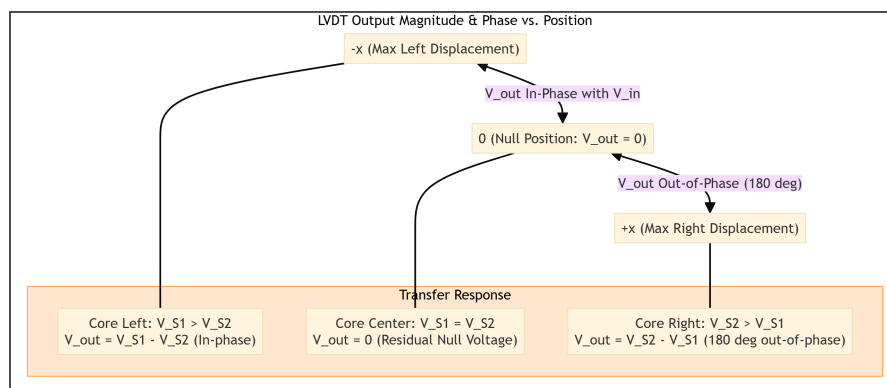


Figure 5.2: LVDT Transfer Characteristics

5.3.3 3. Thermocouple & Cold Junction Compensation (Unit I)

Circuit setup showing measuring and reference junctions with an active compensation bridge circuit.

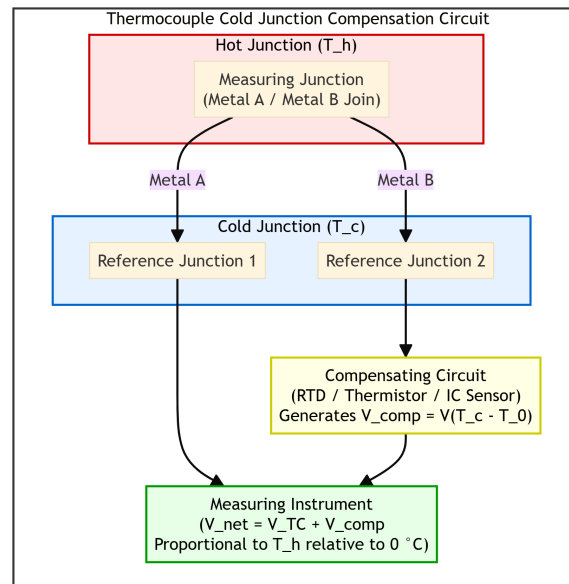


Figure 5.3: Thermocouple & Cold Junction Compensation

5.3.4 4. Pneumatic Spring-Diaphragm Actuator (Unit I)

Mechanical assembly converting control air pressure (3–15 psi) to linear stem displacement.

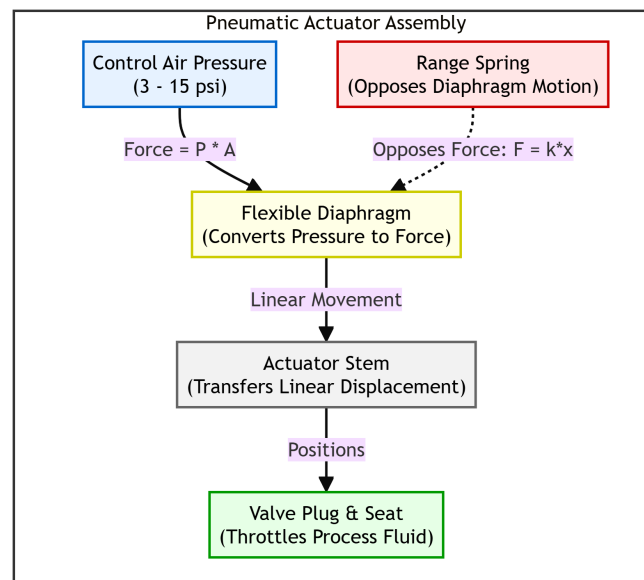


Figure 5.4: Pneumatic Spring-Diaphragm Actuator

5.3.5 5. Signal Conditioning Flow (Unit I)

The sequence of hardware stages processing a physical sensor variable into a digital signal.

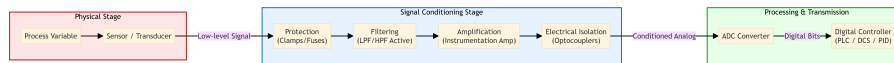


Figure 5.5: Signal Conditioning Functional Architecture

5.3.6 6. Active Filters (Unit I)

Operational amplifier circuits for low-pass and high-pass filtering.

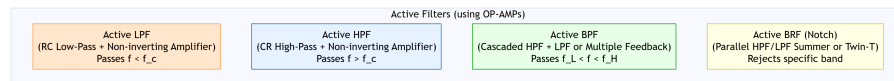


Figure 5.6: Active Filter Circuit Topologies

5.3.7 7. OP-AMP Parallel PID (Unit II)

The physical summing op-amp schematic combining proportional, integral, and derivative paths.

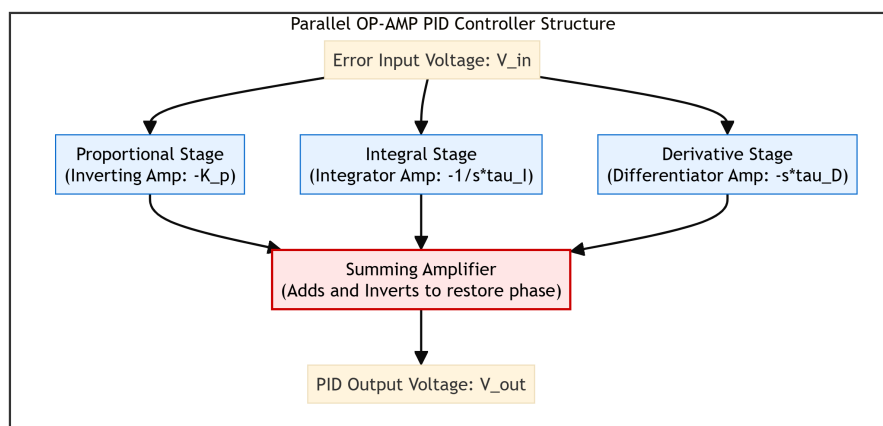


Figure 5.7: Parallel OP-AMP PID Controller

5.3.8 8. PLC Architecture (Unit III)

Internal modules showing CPU, memory, power supply, system bus, and input/output interface cards.

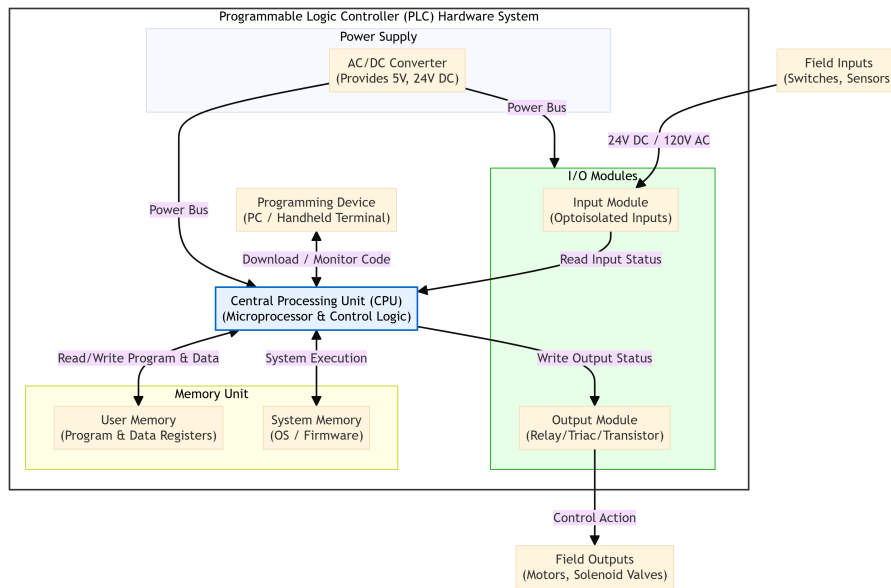


Figure 5.8: PLC Hardware Architecture Block Diagram

5.3.9 9. PLC Ladder Logic (Unit III)

Start/Stop latch logic and Auto-Filling tank logic.

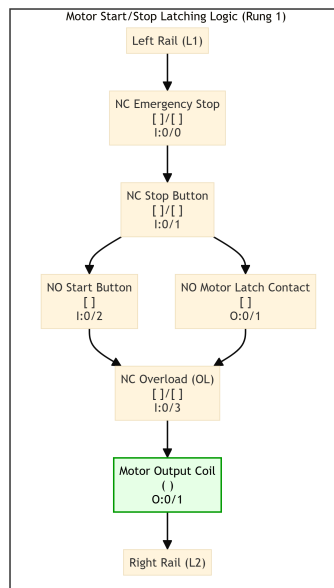


Figure 5.9: Motor Start/Stop Latching Control

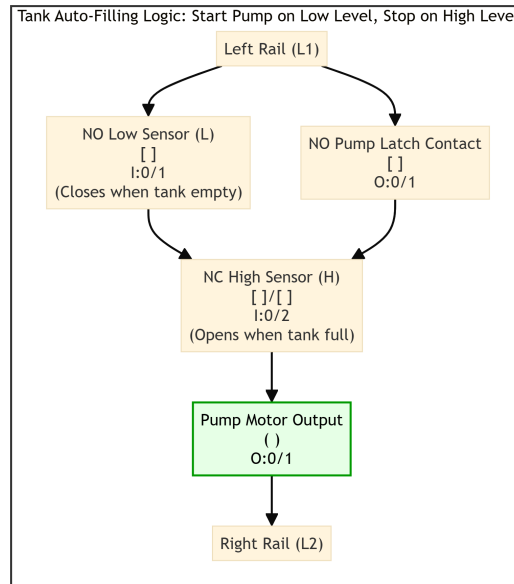


Figure 5.10: PLC Tank Level Control System

5.3.10 10. DCS Levels (Unit III)

The five levels of hierarchical control: Level 0 (Field), Level 1 (Direct Control), Level 2 (Supervisory), Level 3 (Production), and Level 4 (Enterprise).

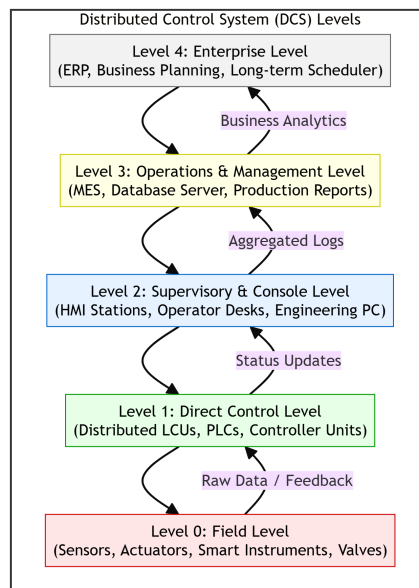


Figure 5.11: DCS Hierarchical Levels

5.3.11 11. SCADA Structure (Unit III)

Geographically distributed control architecture showing the Master Terminal Unit (MTU), Remote Terminal Units (RTUs), PLCs, and WAN link.

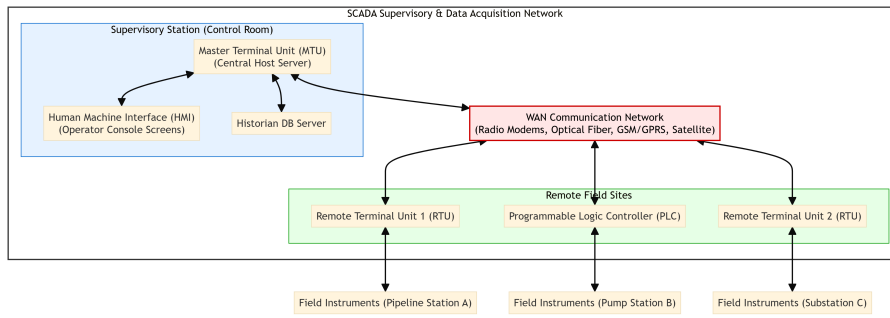


Figure 5.12: SCADA System Topology Block Diagram

5.3.12 12. Cascade Control Loop (Unit IV)

Nested block diagram featuring a primary (outer) loop controller and a fast secondary (inner) loop controller.

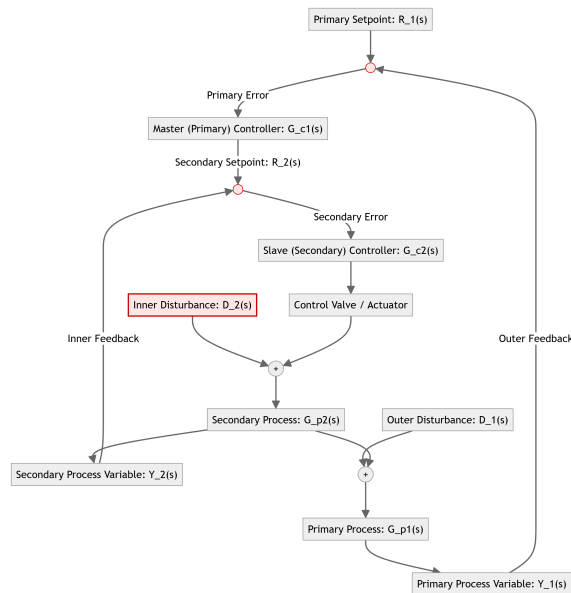


Figure 5.13: Cascade Control Loop Block Diagram

5.3.13 13. Override Control (Unit IV)

Selective control block diagram implementing high/low limit selection to protect hardware.

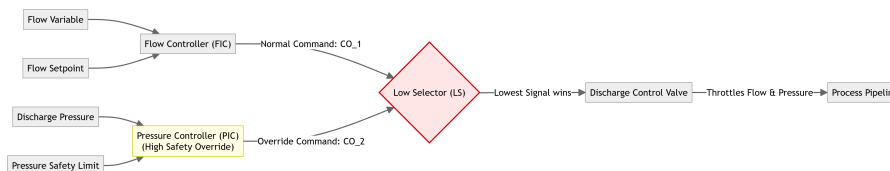


Figure 5.14: Override (Selective) Control Loop

5.3.14 14. Split-Range Control (Unit IV)

Response curves demonstrating a single controller command split to drive multiple actuators sequentially.

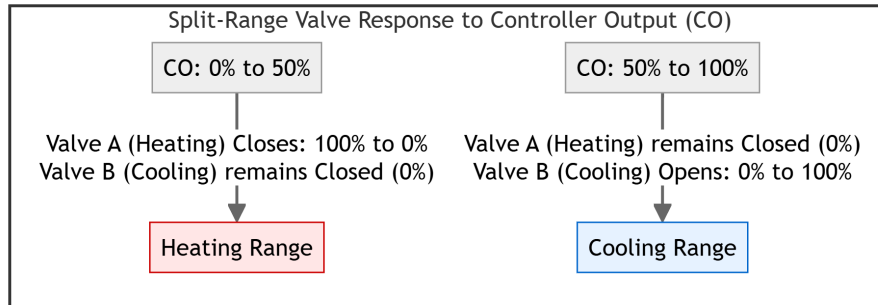


Figure 5.15: Split-Range (Duplex) Control Loop Graph

5.3.15 15. Adaptive Controllers (Unit IV)

Block diagrams for both Model Reference Adaptive Control (MRAC) and Self-Tuning Regulators (STR).

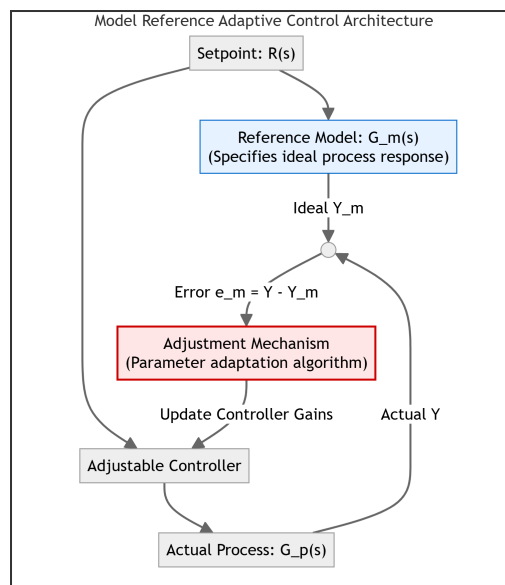


Figure 5.16: Model Reference Adaptive Control (MRAC)

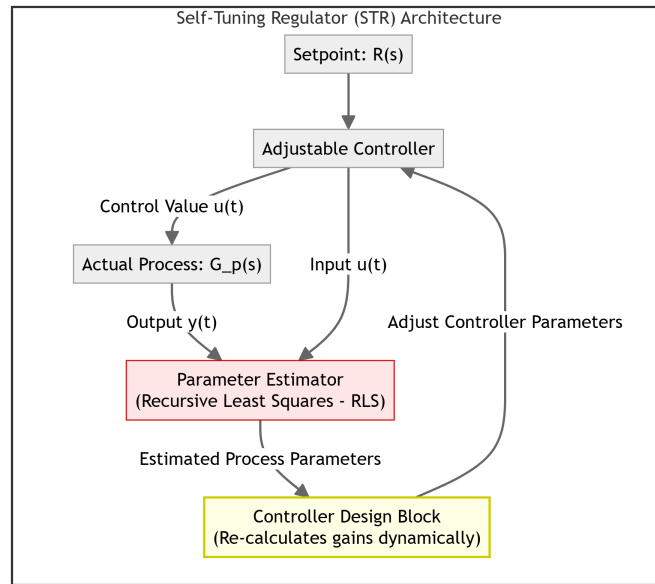


Figure 5.17: Self-Tuning Regulator (STR)

5.3.16 16. IMC Block Diagram (Unit IV)

Internal Model Control block structure feeding back isolated disturbances to the controller.

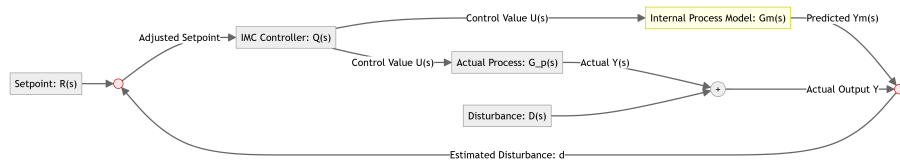


Figure 5.18: Internal Model Control (IMC) Block Diagram

5.4 Section 4: Solved Exam One-Liners (Active Recall Database)

- **Q1.** What is the operating frequency of an ultrasonic sensor?
- *Answer:* **10–65 kHz**
- **Q2.** The primary sensing element of the pressure thermometer is
- *Answer:* **Pressure bulb filled with a liquid**
- **Q3.** Thermocouples generate output voltage according to the
- *Answer:* **Seebeck effect**
- **Q4.** Touch screen devices use which sensor?
- *Answer:* **Capacitive sensors**
- **Q5.** What is the purpose of a pressure gauge in a boiler plant?
- *Answer:* **To measure pressure and display it**

- **Q6. State the cause for using starters with D.C. motors.**
- *Answer: To limit the high starting current*
- **Q7. The span of the pneumatic signal used in industry is**
- *Answer: 3-15 psi*
- **Q8. The gauge factor of a strain gauge indicates its**
- *Answer: Sensitivity*
- **Q9. Cohen-Coon tuning rules are used for systems.**
- *Answer: Processes with time-lag and self-regulation*
- **Q10. What type of controller is used for the elimination of offset?**
- *Answer: Integral controller (I-controller)*
- **Q11. The Ziegler-Nichols tuning method is used for**
- *Answer: Processes with time-lag and self-regulation*
- **Q12. The derivative mode (D controller) is also known as**
- *Answer: Rate controller*
- **Q13. If the proportional band of an electronic PID controller is set at 10, then what is the proportional gain (K_c)?**
- *Answer: Proportional gain $K_c = 100/PB = 100/10 = 10$*
- **Q14. The quarter-amplitude decay ratio is basically a design criterion specified by the Ziegler-Nichols method which implies that the amplitude of an oscillation must be reduced by a factor of**
- *Answer: 1/4*
- **Q15. A summing amplifier block followed by an inverter block represents a controller.**
- *Answer: PI/PID controller (when integrating and proportional resistors are configured)*
- **Q16. What are the advantages of a pneumatic controller?**
- *Answer: Simple construction, explosion-proof/safe in hazardous areas, high power output*
- **Q17. Which of the following controllers has the maximum offset? (a) P-controller (b) P-I controller (c) P-I-D controller (d) P-D controller**
- *Answer: (a) P-controller*
- **Q18. In an electronic Integral controller, if the resistance and capacitance values are both doubled, then the controller response time will be and the controller gain will be**

- *Answer: Doubled, quartered* (since integration time constant $\tau_I = RC$ is quadrupled)
- **Q19. If the Proportional Band (PB) of a P-controller is very wide, then the system response is _____.**
- *Answer: Slow (and has high offset)*
- **Q20. The function of reset action in a process controller is to _____.**
- *Answer: Eliminate offset (steady-state error)*
- **Q21. Which controllers do not show a continuous action?**
- *Answer: Digital controllers (discrete-time action)*
- **Q22. What is the transfer function of a dead-time element?**
- *Answer: $G(s) = e^{-sT_d}$*
- **Q23. Provide an example of a self-regulating system.**
- *Answer: Liquid level in a tank with constant gravity outflow*
- **Q24. Which control valve is used for pressure control?**
- *Answer: Globe valve (often used for throttling control)*
- **Q25. Database management and decisions regarding production are handled in which level of the DCS structure?**
- *Answer: Level 3 (Production control/Management level)*
- **Q26. What are the components of a SCADA system?**
- *Answer: MTU (Master Terminal Unit), RTU (Remote Terminal Unit) or PLCs, Communication Network, HMI (Human Machine Interface)*
- **Q27. When a control valve plug is fully connected to the _____, then it is fully closed.**
- *Answer: Valve seat*
- **Q28. A cascade controller is used when the process shows _____.**
- *Answer: Slow response and large time lag (multiple capacities)*
- **Q29. State the load disturbances in the feed-forward control strategy of a simple heat exchanger.**
- *Answer: Inlet flow rate, inlet temperature, and fluid composition*
- **Q30. Which control scheme is known as anticipatory control?**
- *Answer: Feed-forward control*
- **Q31. When should Cascade Control be used?**

- *Answer:* When there are significant secondary disturbances that affect the primary loop
- **Q32.** The response of Feed-forward control is _____ than Feedback control.
- *Answer:* Faster (proactive/anticipatory)
- **Q33.** A ratio control system is a special type of _____ control system.
- *Answer:* Feed-forward control system
- **Q34.** The purpose of ratio control is to _____.
- *Answer:* Maintain a constant ratio between two process variables (e.g., fuel and air ratio)
- **Q35.** LVDT works on the principle of _____.
- *Answer:* Mutual inductance
- **Q36.** What does RFI stand for in industrial instrumentation electrical noise?
- *Answer:* Radio Frequency Interference
- **Q37.** The device which converts a 4–20 mA current loop signal to a 3–15 psi pneumatic signal is called: _____.
- *Answer:* Electronic-to-Pneumatic (I/P) Converter
- **Q38.** Digital control systems are generally superior to analog control systems, but they are always worse with regard to: _____.
- *Answer:* Noise immunity
- **Q39.** In a PLC, the scan time refers to the amount of time in which: _____.
- *Answer:* The entire program takes to execute
- **Q40.** What is a P to I converter in process control?
- *Answer:* Pneumatic to Current (P/I) Converter
- **Q41.** What is the standard SI unit of torque?
- *Answer:* Newton meter (N·m)
- **Q42.** Integral control action in a closed-loop system acts to: _____.
- *Answer:* Decrease the steady-state error (eliminating offset)
- **Q43.** For a fast pressure control loop with a PI controller and a process time constant of 10 seconds, the best actuator choice is: _____.
- *Answer:* A piston and cylinder actuator
- **Q44.** In a direct acting controller (using the CA4 key definition), the output increases when the process measurement: _____.

- **Answer: Decreases** (*Note: In standard control theory, a direct-acting controller's output increases when the process measurement increases; if error is defined as $e = SP - PV$, then output increases as PV decreases*).
- **Q45. A PID controller is preferred when a process loop requires: _____.**
- **Answer: Offset must be eliminated, process variations are small, and fast recovery is needed**
- **Q46. A capacitive displacement sensor can measure very small displacements by varying: _____.**
- **Answer: Either plate separation, overlapping area, or dielectric permittivity**
- **Q47. What is the minimum number of operational amplifiers required to construct an electronic parallel PID controller?**
- **Answer: 2 operational amplifiers**
- **Q48. The instruction used to turn an output ON or OFF after its input has been ON for a preset duration is: _____.**
- **Answer: Timer On Delay (TON)**
- **Q49. A potentiometer is primarily used for the measurement of: _____.**
- **Answer: Linear displacement** (rotary types measure angular displacement)
- **Q50. Control valve sizing is required in engineering design to avoid: _____.**
- **Answer: Undersizing, oversizing, and excess cost (All of the above)**
- **Q51. The Proportional Band (PB) of a process controller is expressed in what unit?**
- **Answer: Percentage** ($PB = 100/K_c$)
- **Q52. The output signal of smart sensors is: _____.**
- **Answer: Digital**
- **Q53. The Ziegler-Nichols tuning method is: _____.**
- **Answer: A graphical solution method that can determine controller gains for a PI/PID loop**
- **Q54. The noise immunity of a PLC to electrical noise as compared to conventional relay controllers is: _____.**
- **Answer: Excellent**
- **Q55. An automatic kitchen toaster is an example of which loop type?**
- **Answer: Open-loop control system**
- **Q56. Which components are required to form a standard 4–20 mA current loop?**

- *Answer:* A DC power supply, wire, and a current transmitter (All of the above)
- **Q57. If the overshoots of a PID control loop have increased, the derivative time constant (T_d) must be: -----.**
- *Answer:* **Increased**
- **Q58. Which of the following is NOT an industrial PLC manufacturer?**
- *Answer:* **Microsoft** (Major manufacturers include Siemens, Allen-Bradley, Mitsubishi, ABB, and Schneider Electric)